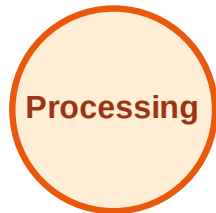


# BFS Traversal

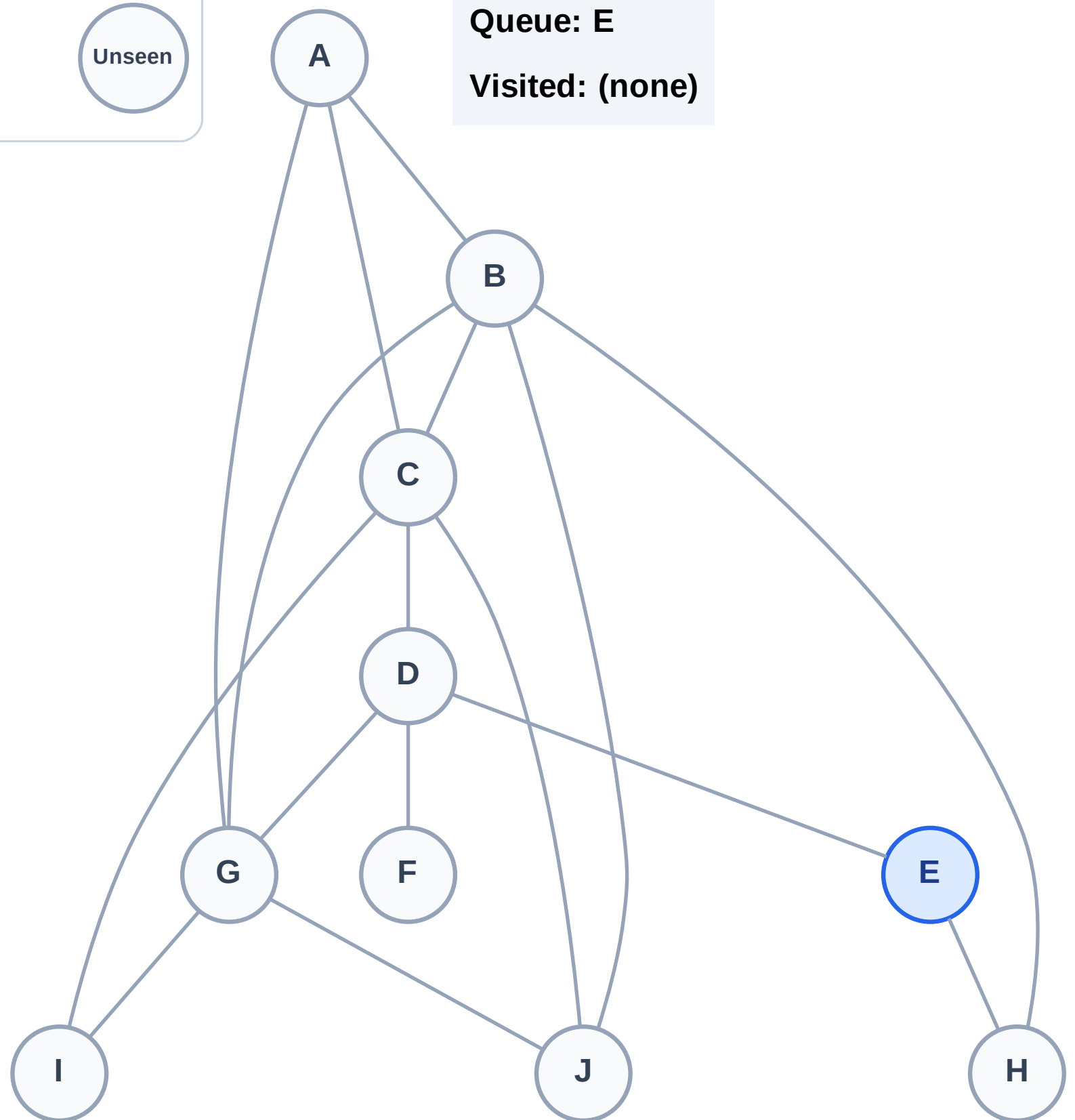
Step 1: Start at E

## Legend



Queue: E

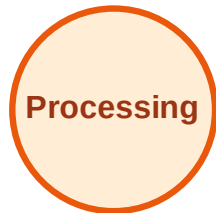
Visited: (none)



# BFS Traversal

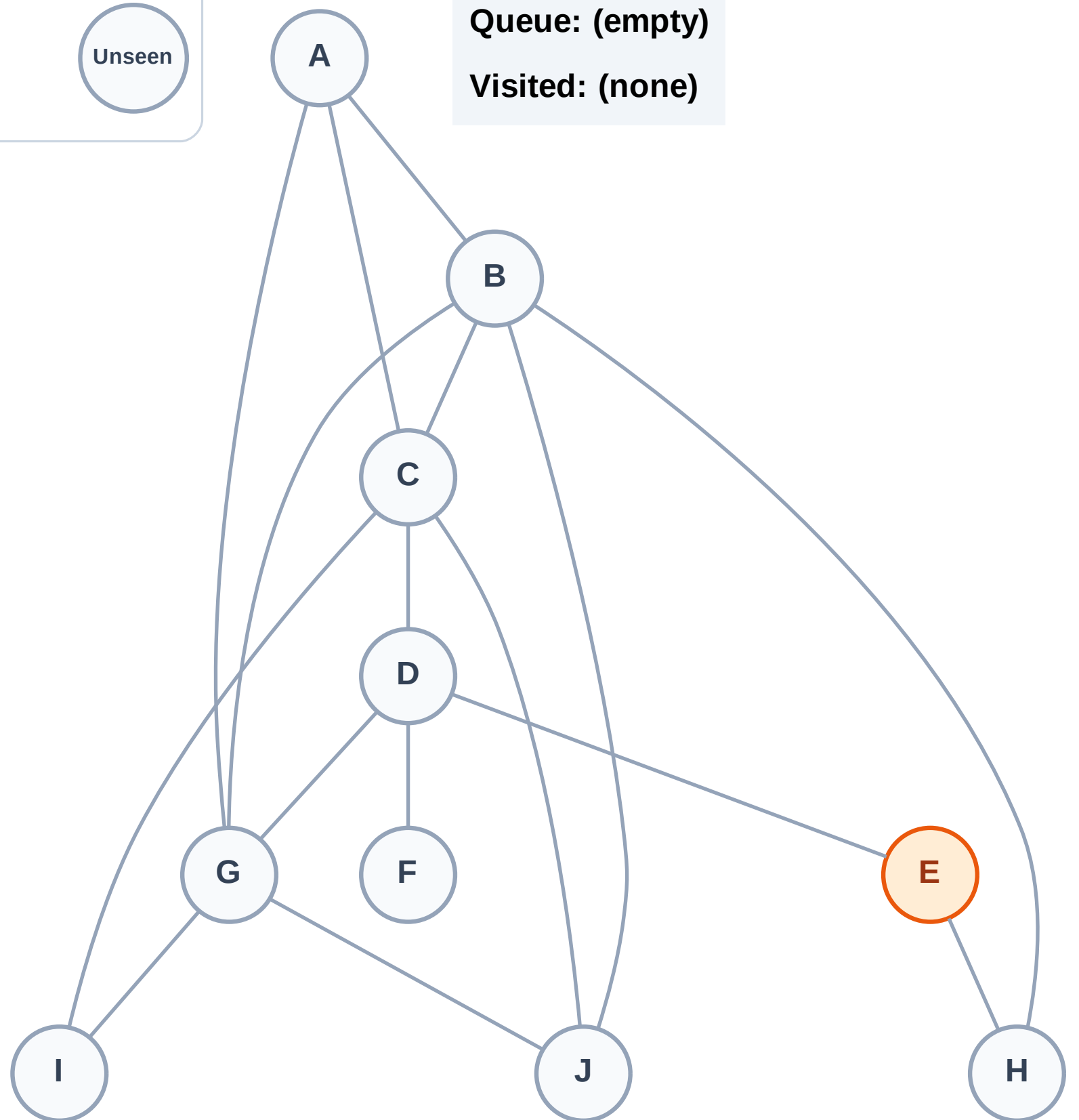
Step 2: Process node E

## Legend



Queue: (empty)

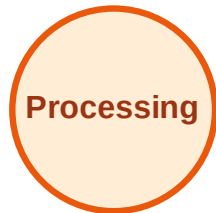
Visited: (none)



# BFS Traversal

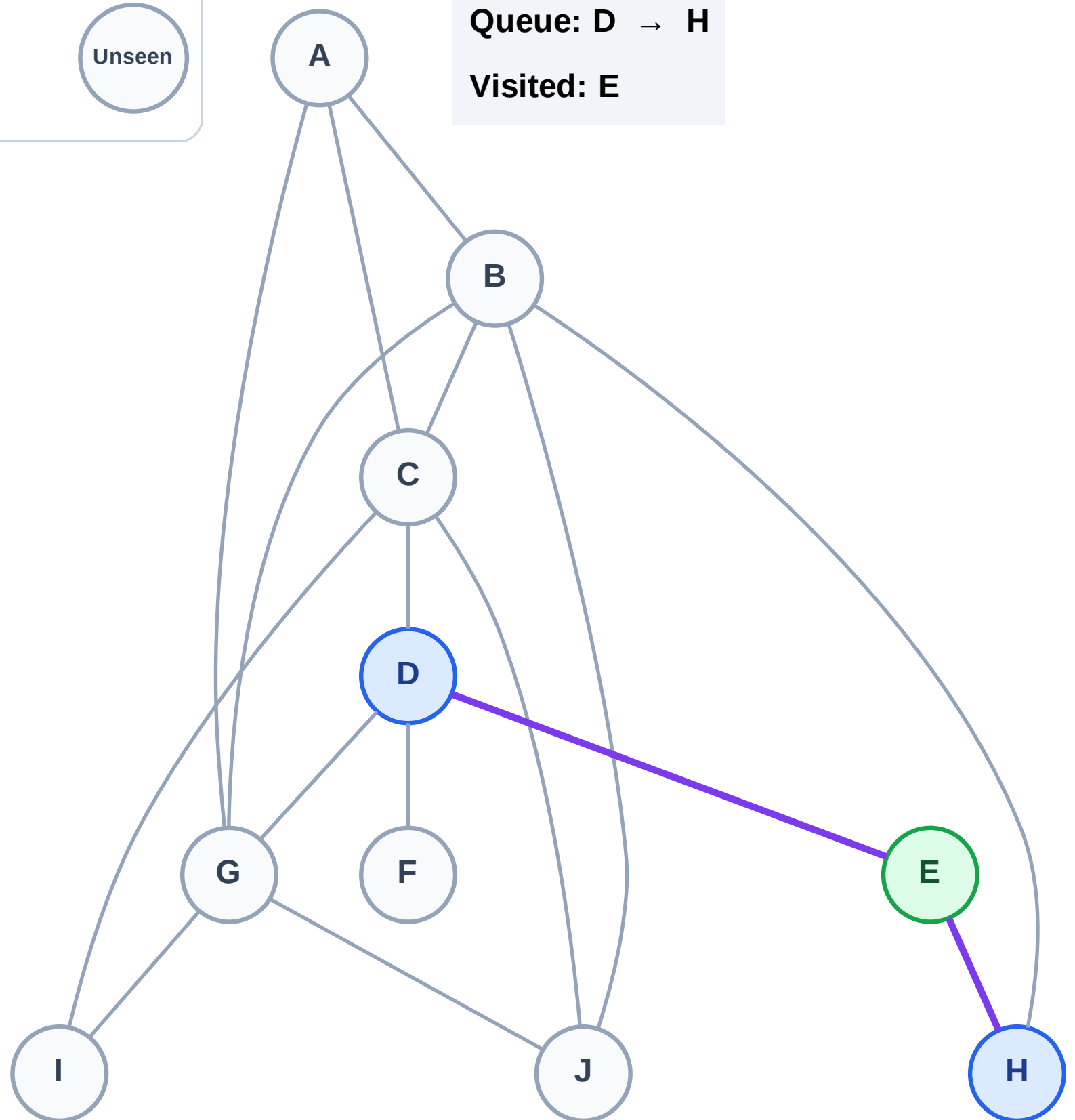
Step 3: Discover D, H from E

## Legend



Queue: D → H

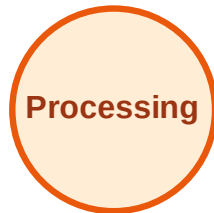
Visited: E



# BFS Traversal

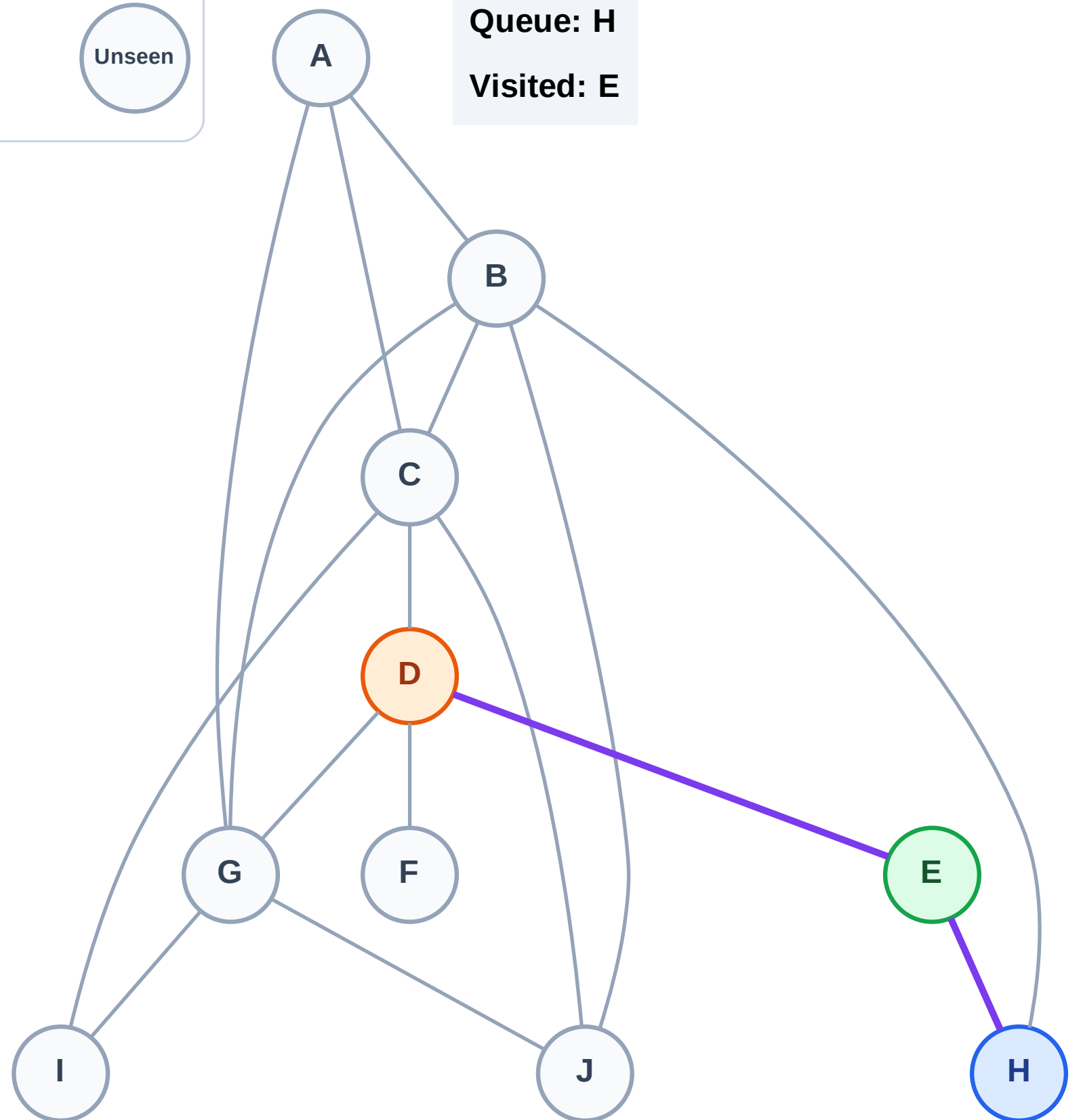
Step 4: Process node D

## Legend



Queue: H

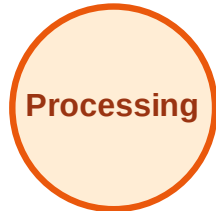
Visited: E



# BFS Traversal

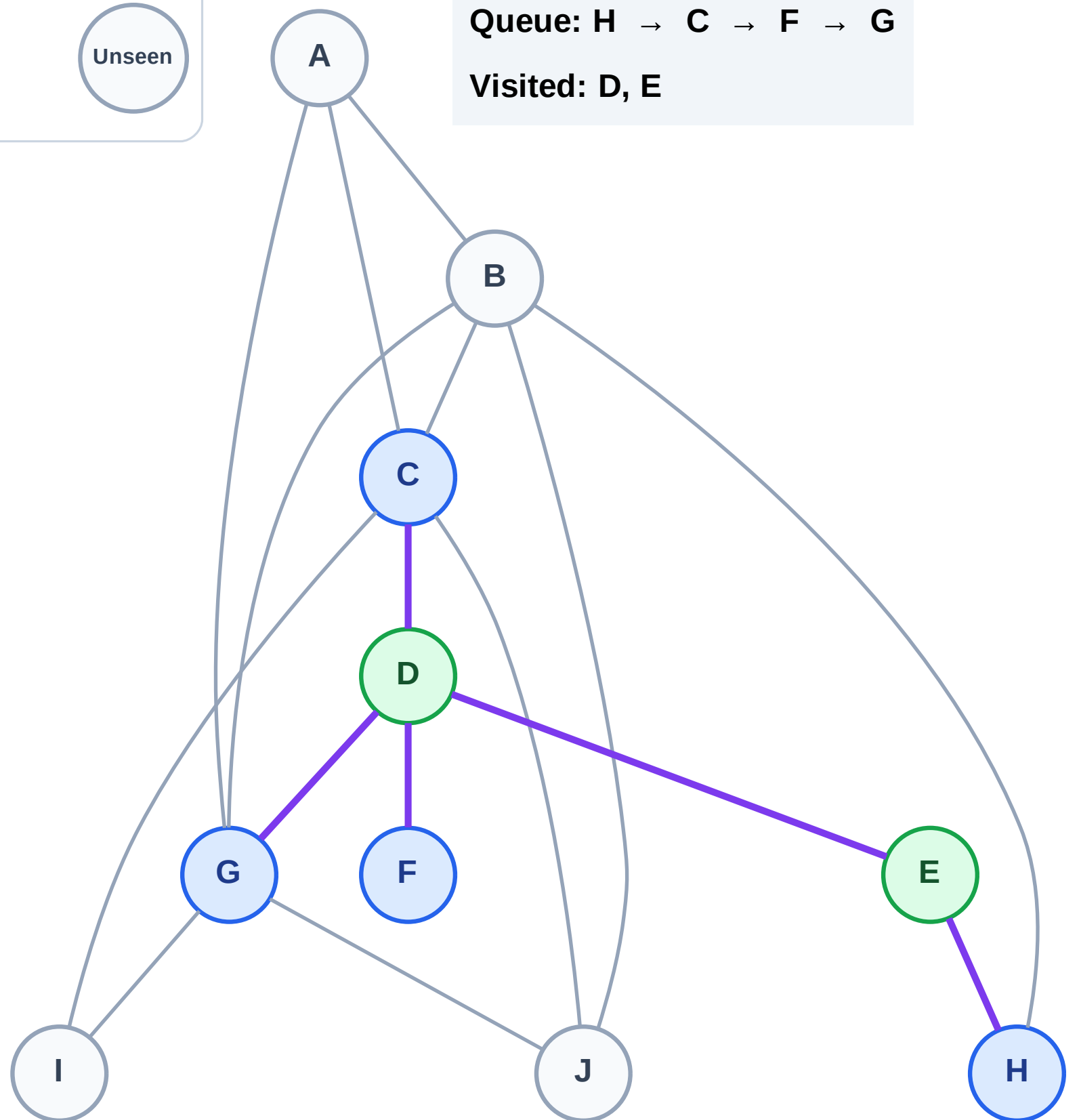
Step 5: Discover C, F, G from D

## Legend



Queue: H → C → F → G

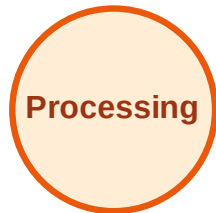
Visited: D, E



# BFS Traversal

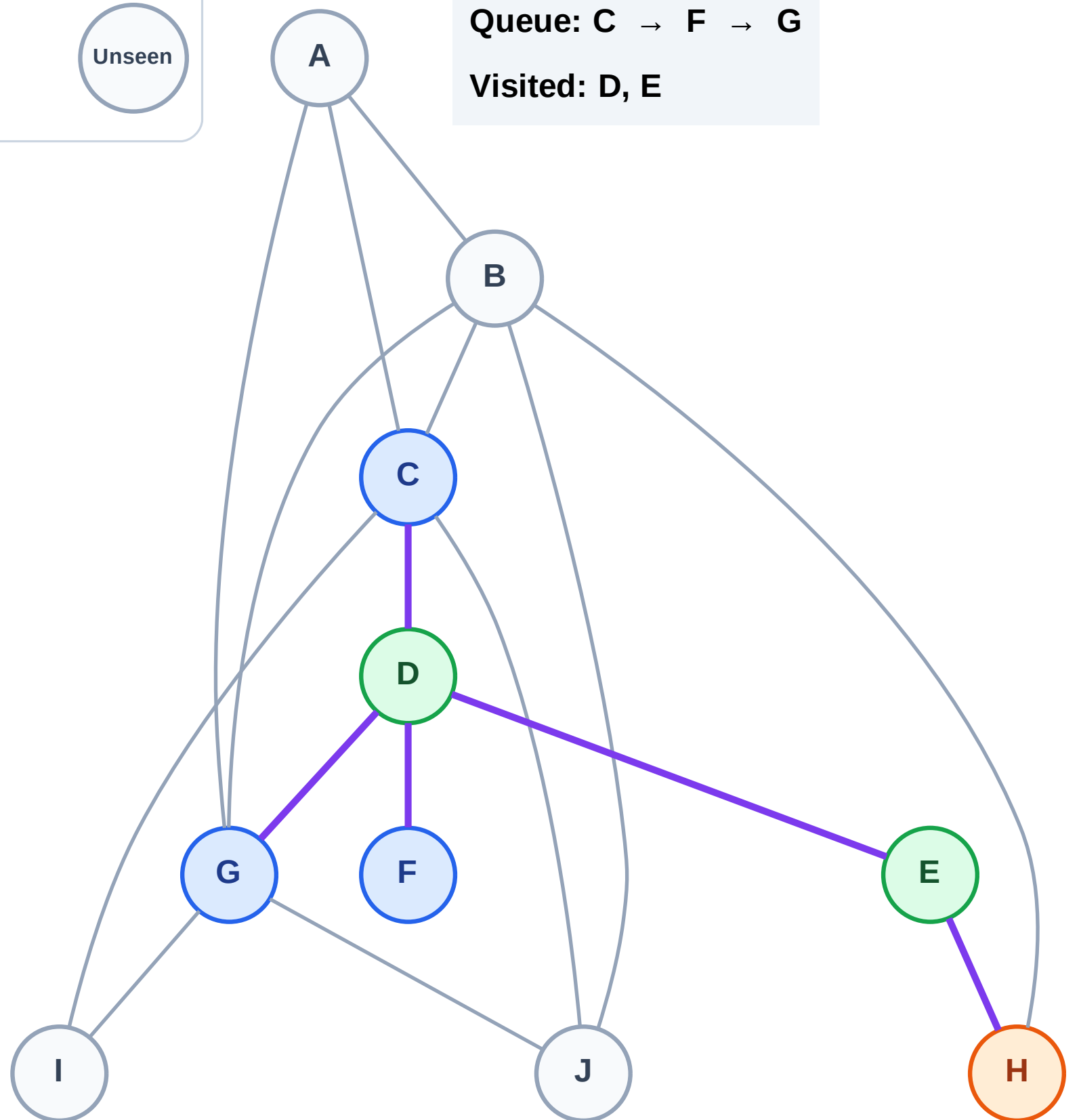
Step 6: Process node H

## Legend



Queue: C → F → G

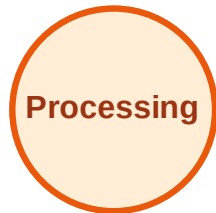
Visited: D, E



# BFS Traversal

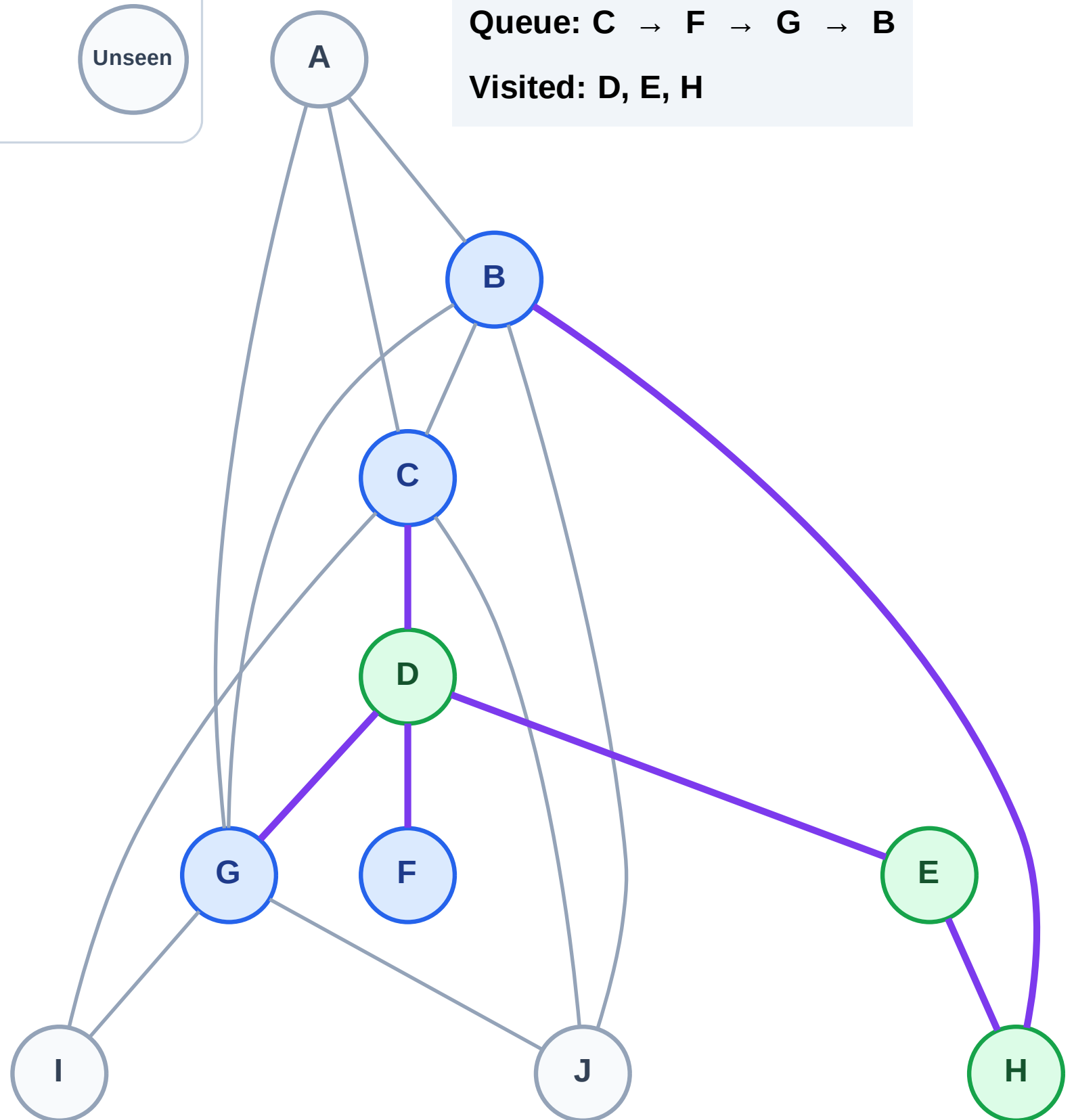
Step 7: Discover B from H

## Legend



Queue: C → F → G → B

Visited: D, E, H

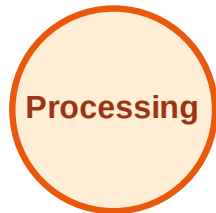




# BFS Traversal

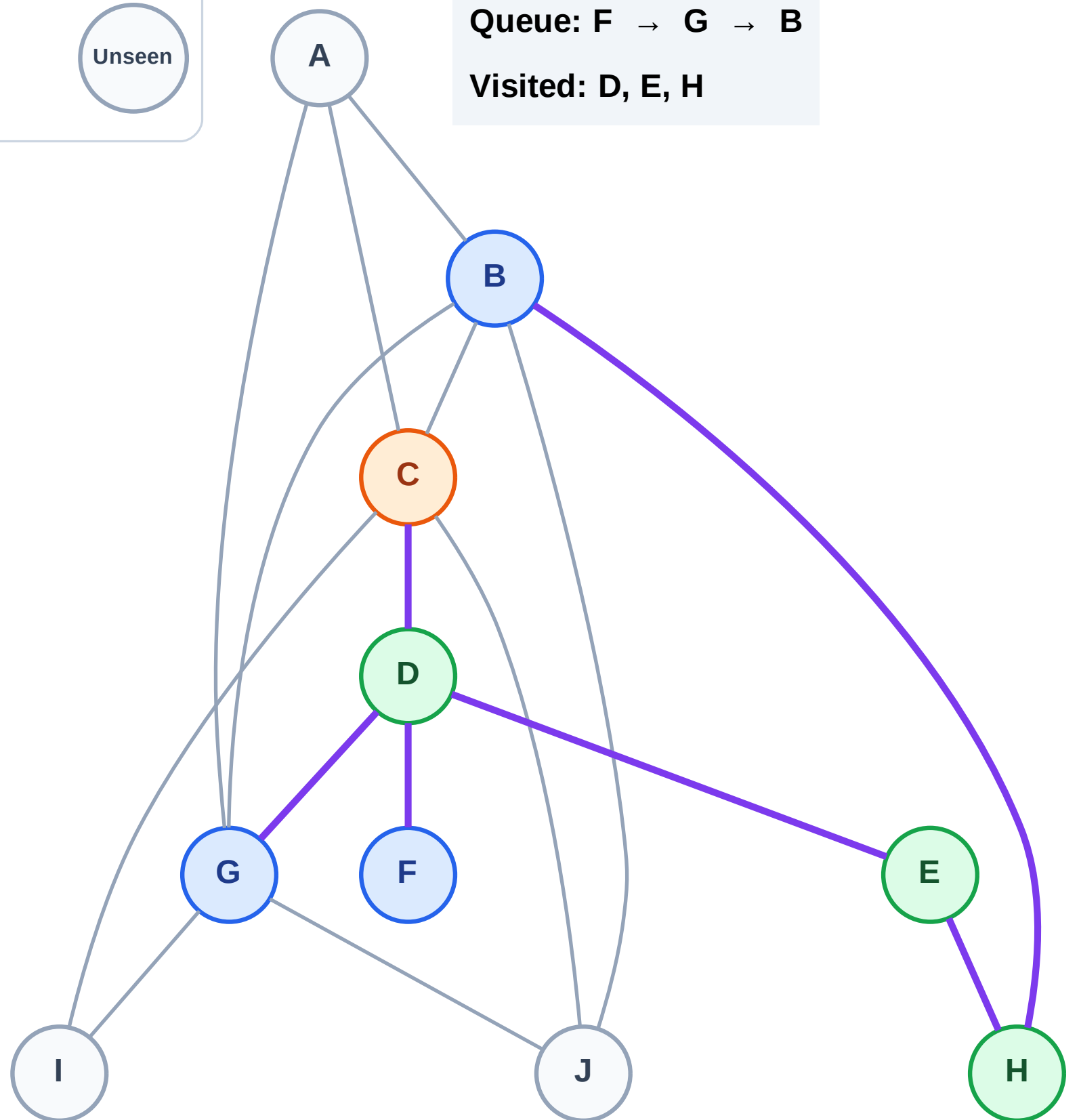
Step 8: Process node C

## Legend



Queue: F → G → B

Visited: D, E, H

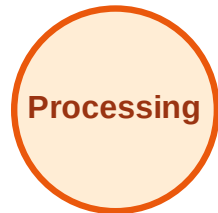




# BFS Traversal

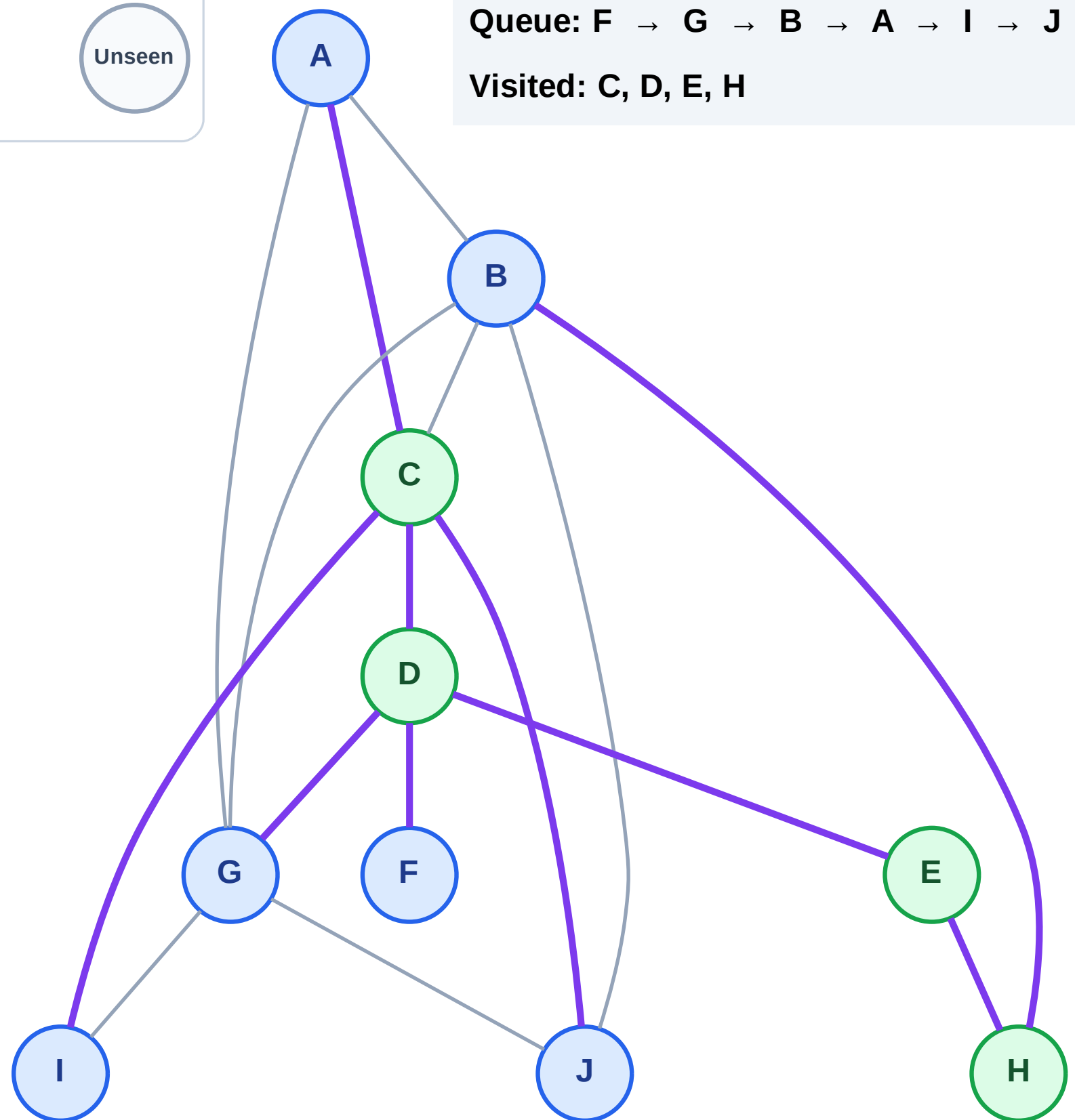
Step 9: Discover A, I, J from C

## Legend



Queue: F → G → B → A → I → J

Visited: C, D, E, H



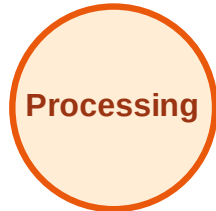
# BFS Traversal

Step 10: Process node F

## Legend



Complete



Processing



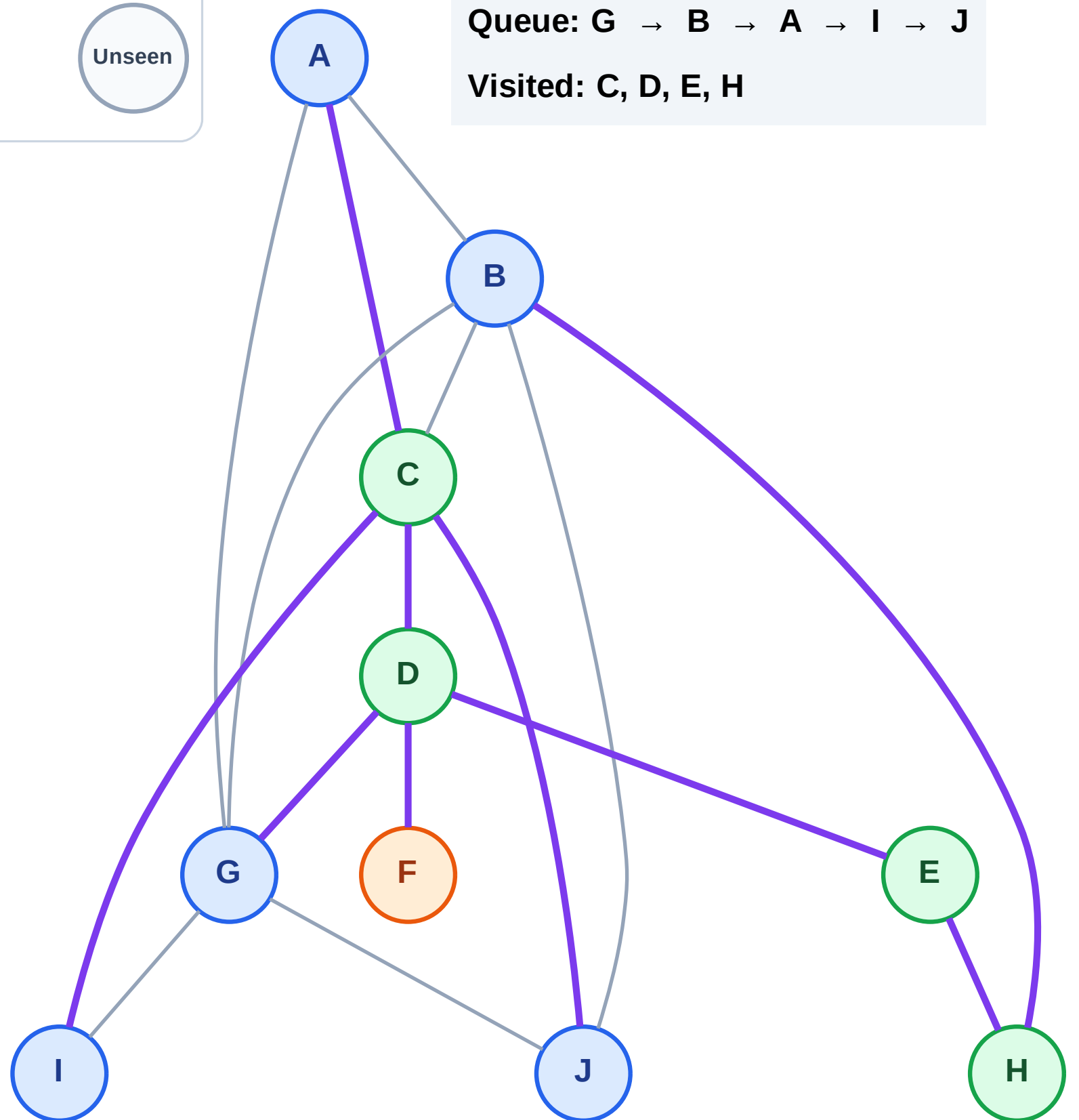
Discovered



Unseen

Queue: G → B → A → I → J

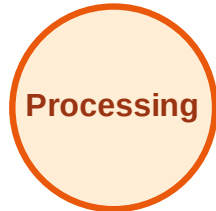
Visited: C, D, E, H



# BFS Traversal

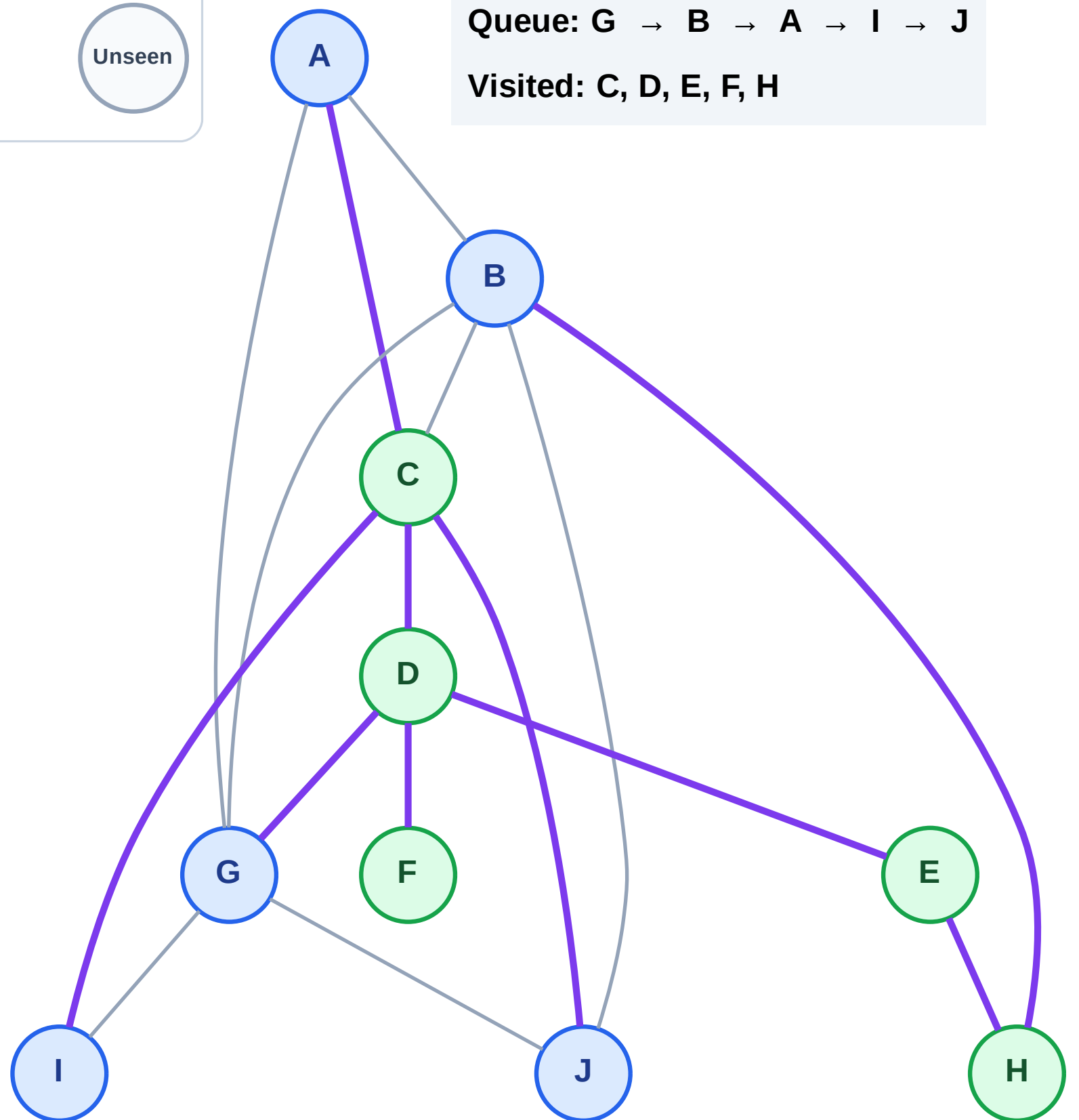
Step 11: No new neighbors from F

## Legend



Queue: G → B → A → I → J

Visited: C, D, E, F, H



# BFS Traversal

Step 12: Process node G

## Legend

Complete

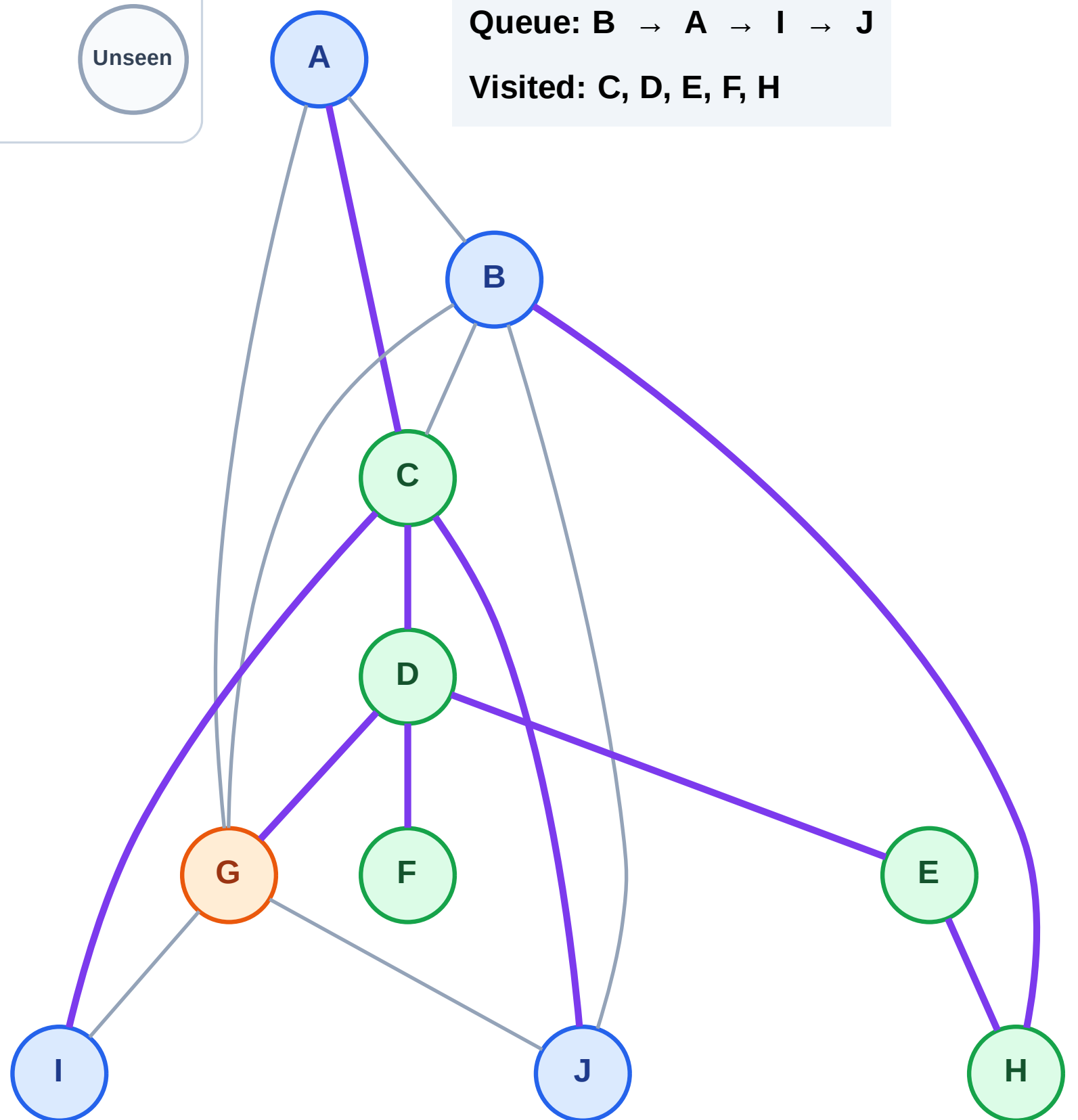
Processing

Discovered

Unseen

Queue: B → A → I → J

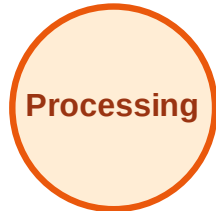
Visited: C, D, E, F, H



# BFS Traversal

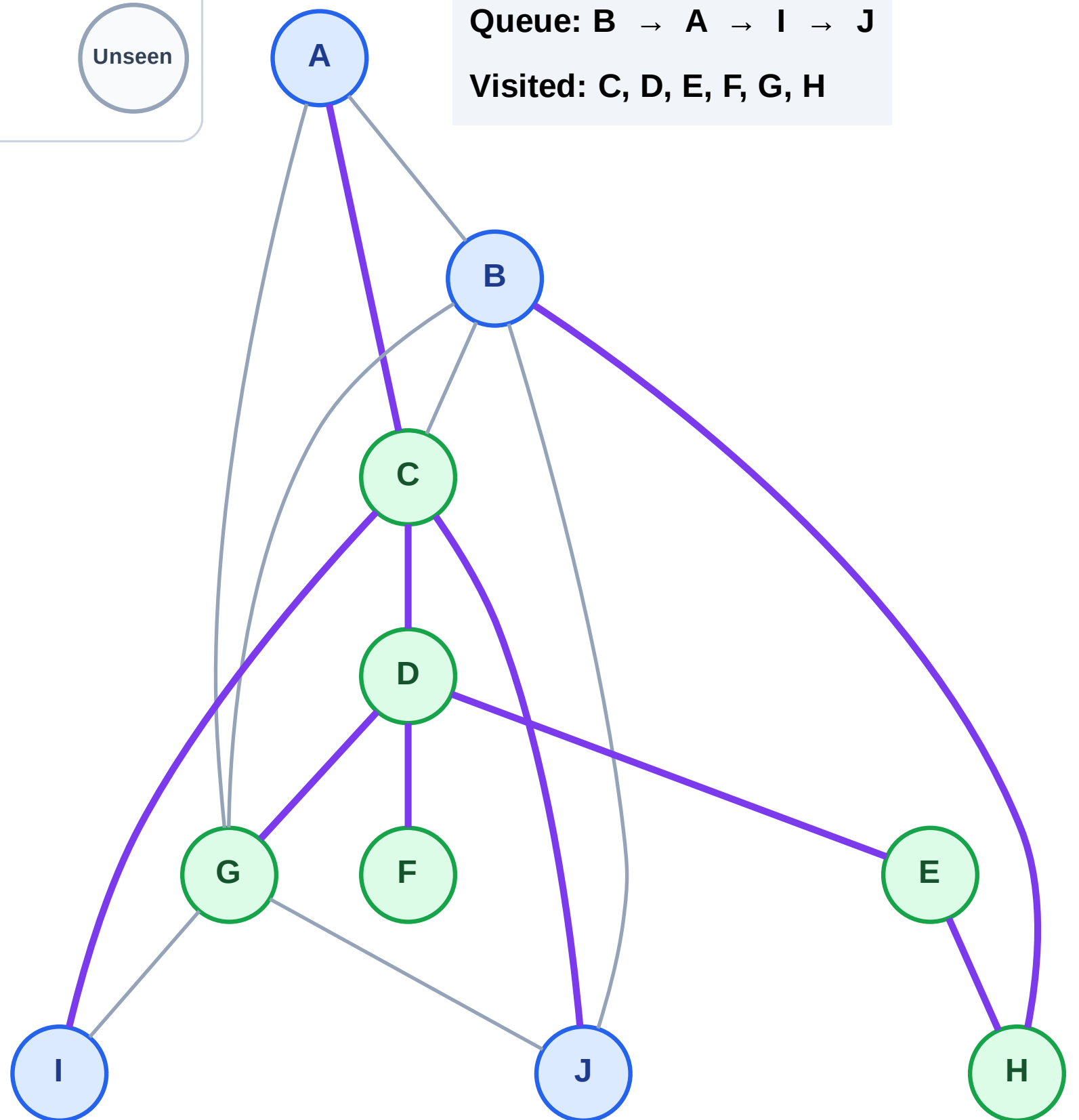
Step 13: No new neighbors from G

## Legend



Queue: B → A → I → J

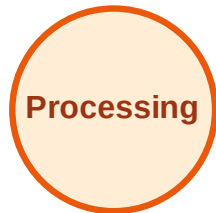
Visited: C, D, E, F, G, H



# BFS Traversal

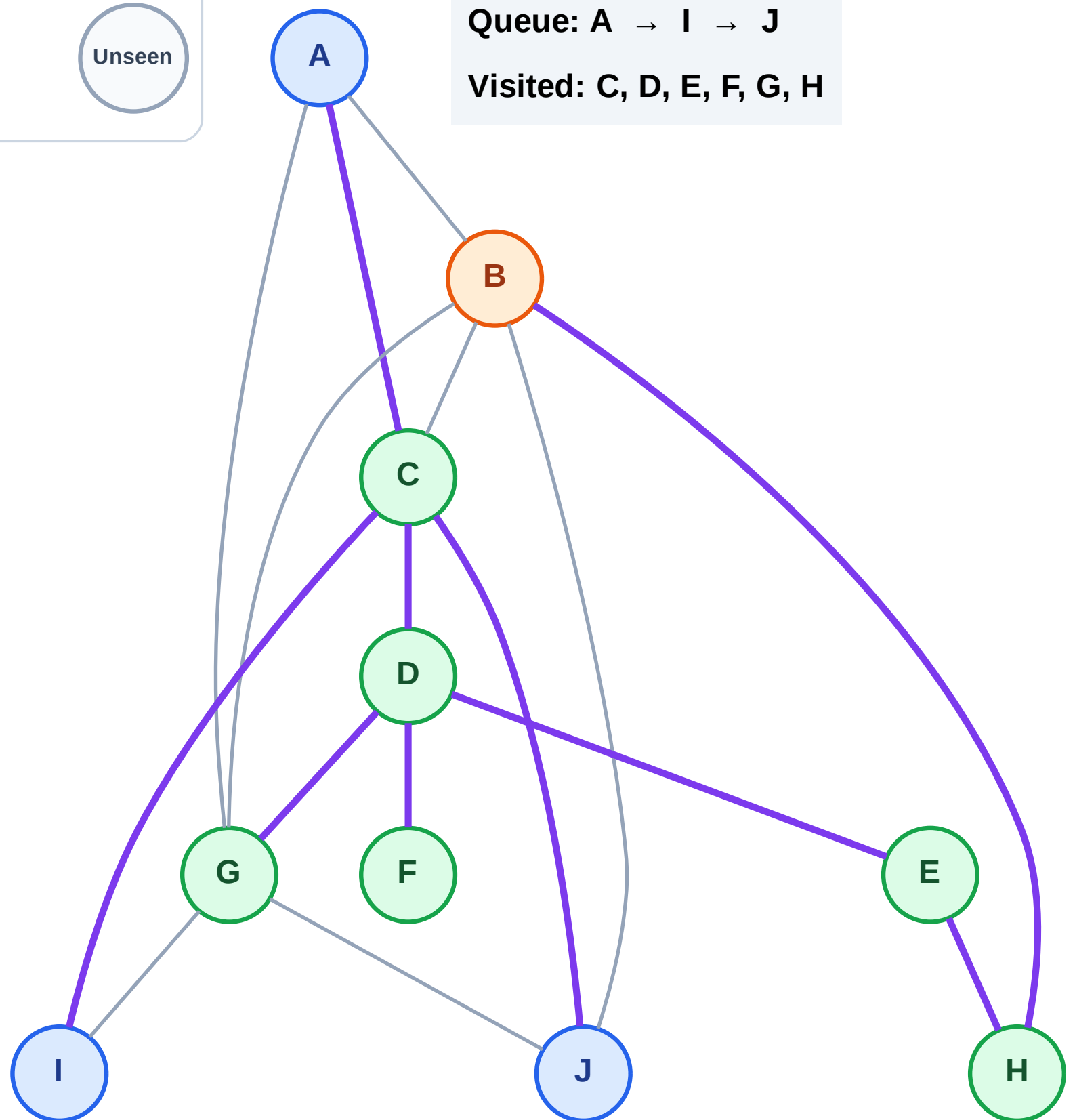
Step 14: Process node B

## Legend



Queue: A → I → J

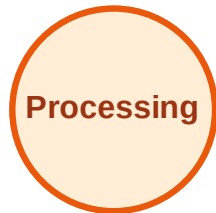
Visited: C, D, E, F, G, H



# BFS Traversal

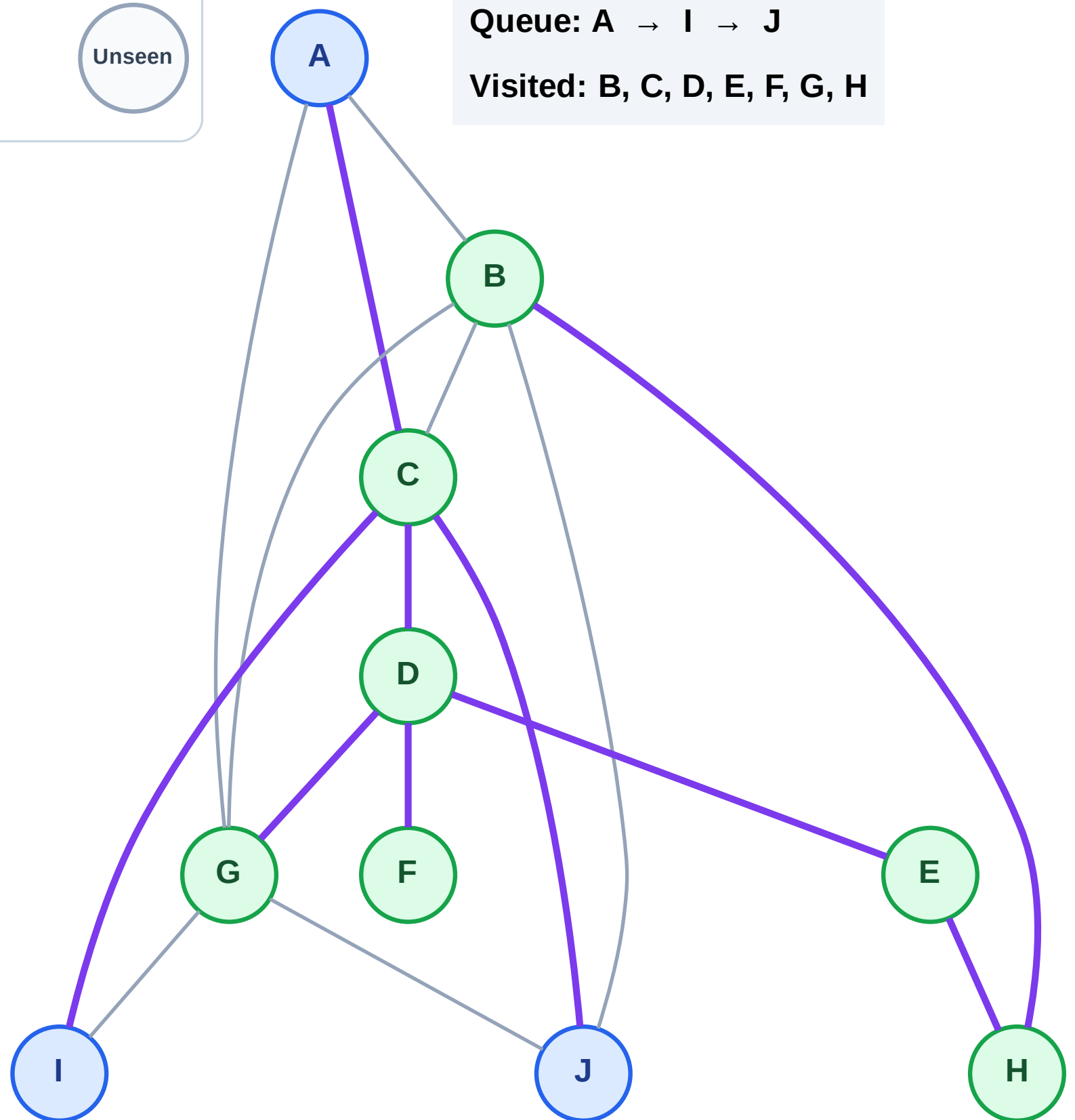
Step 15: No new neighbors from B

## Legend



Queue: A → I → J

Visited: B, C, D, E, F, G, H

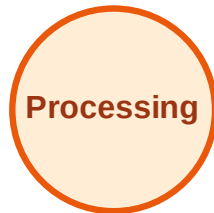




# BFS Traversal

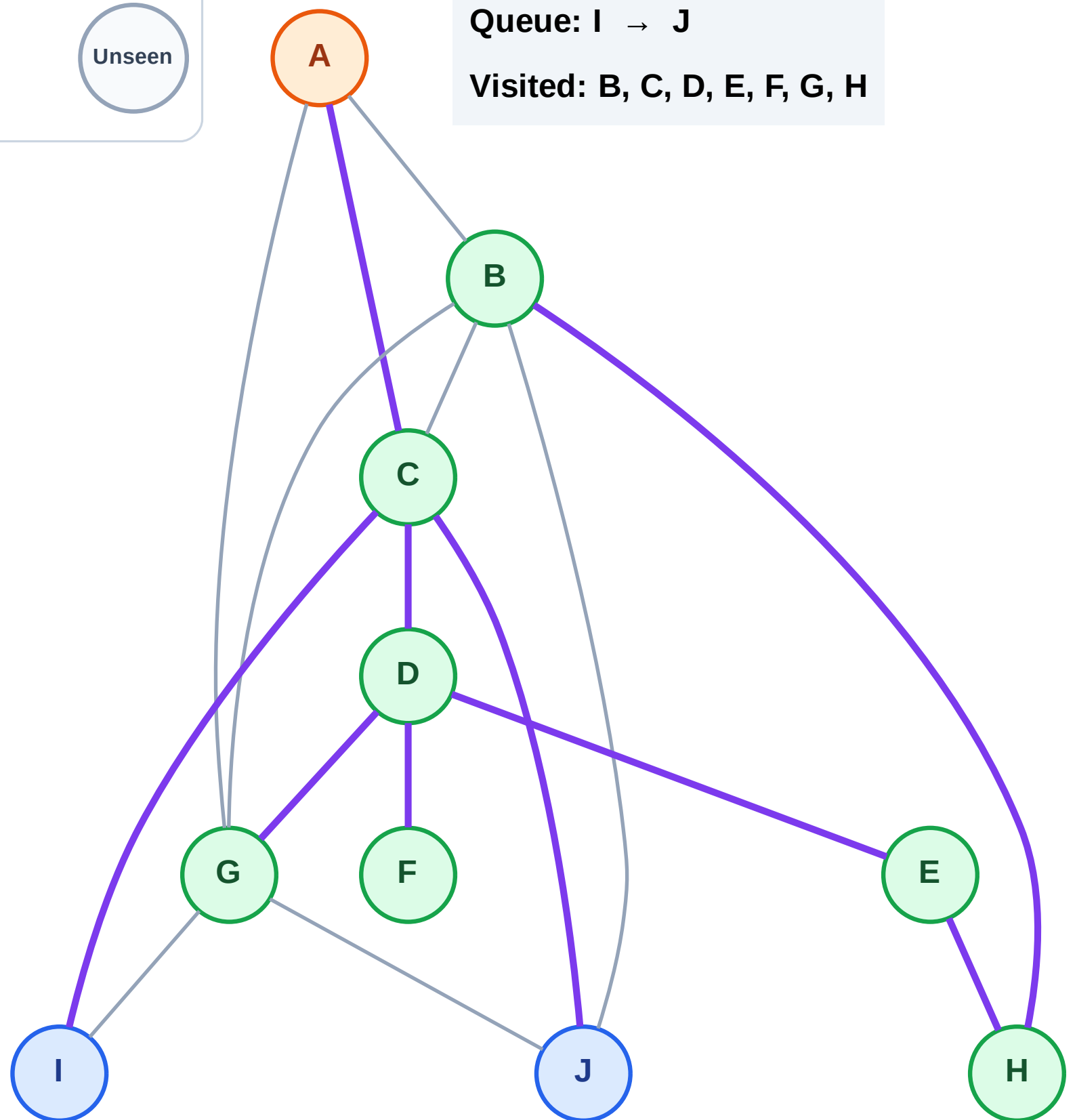
Step 16: Process node A

## Legend



Queue: I → J

Visited: B, C, D, E, F, G, H

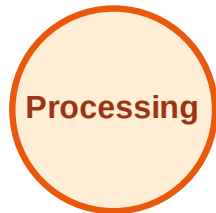




# BFS Traversal

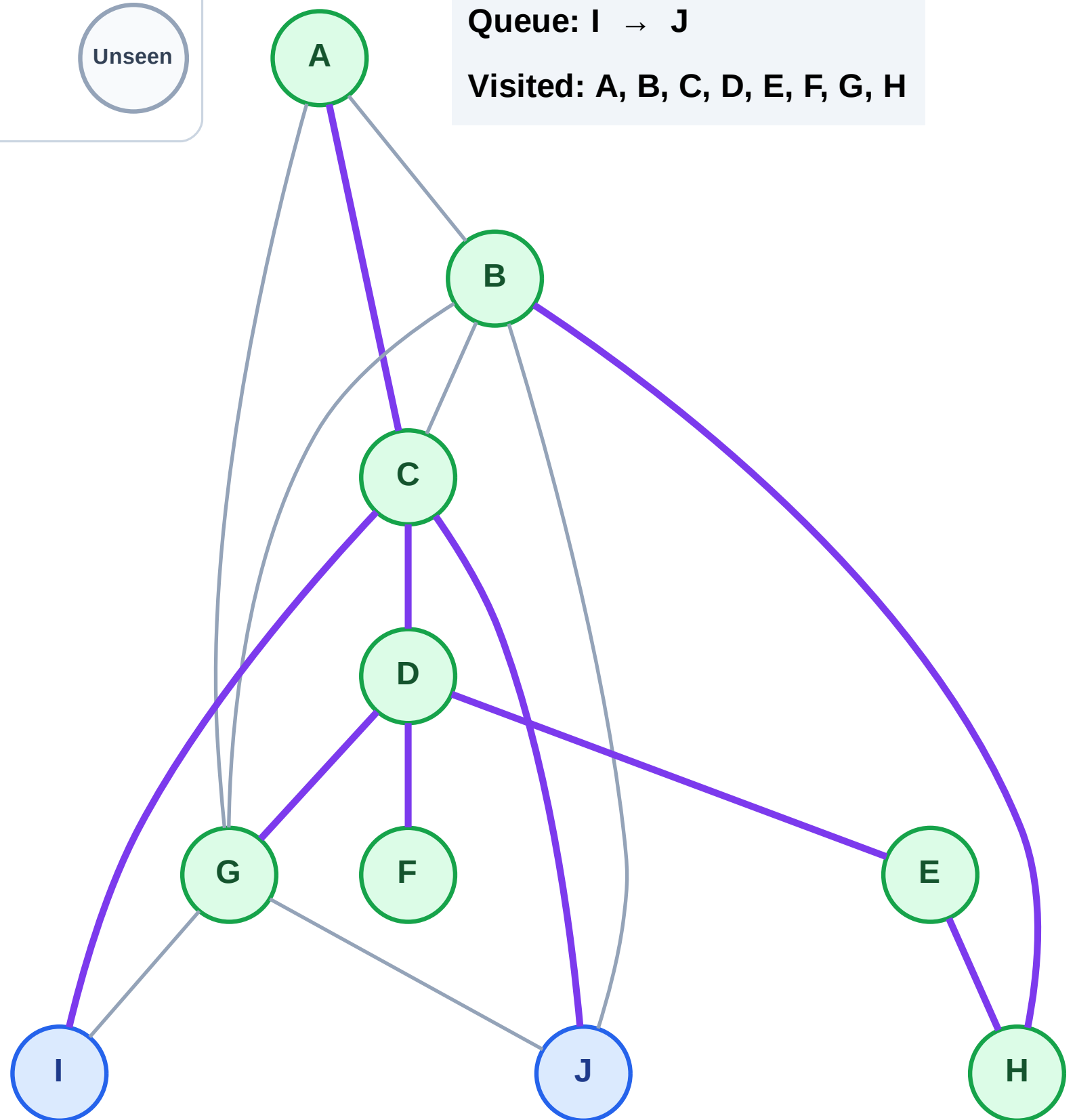
Step 17: No new neighbors from A

## Legend



Queue: I → J

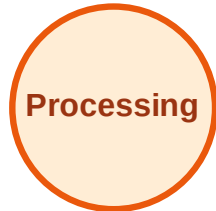
Visited: A, B, C, D, E, F, G, H



# BFS Traversal

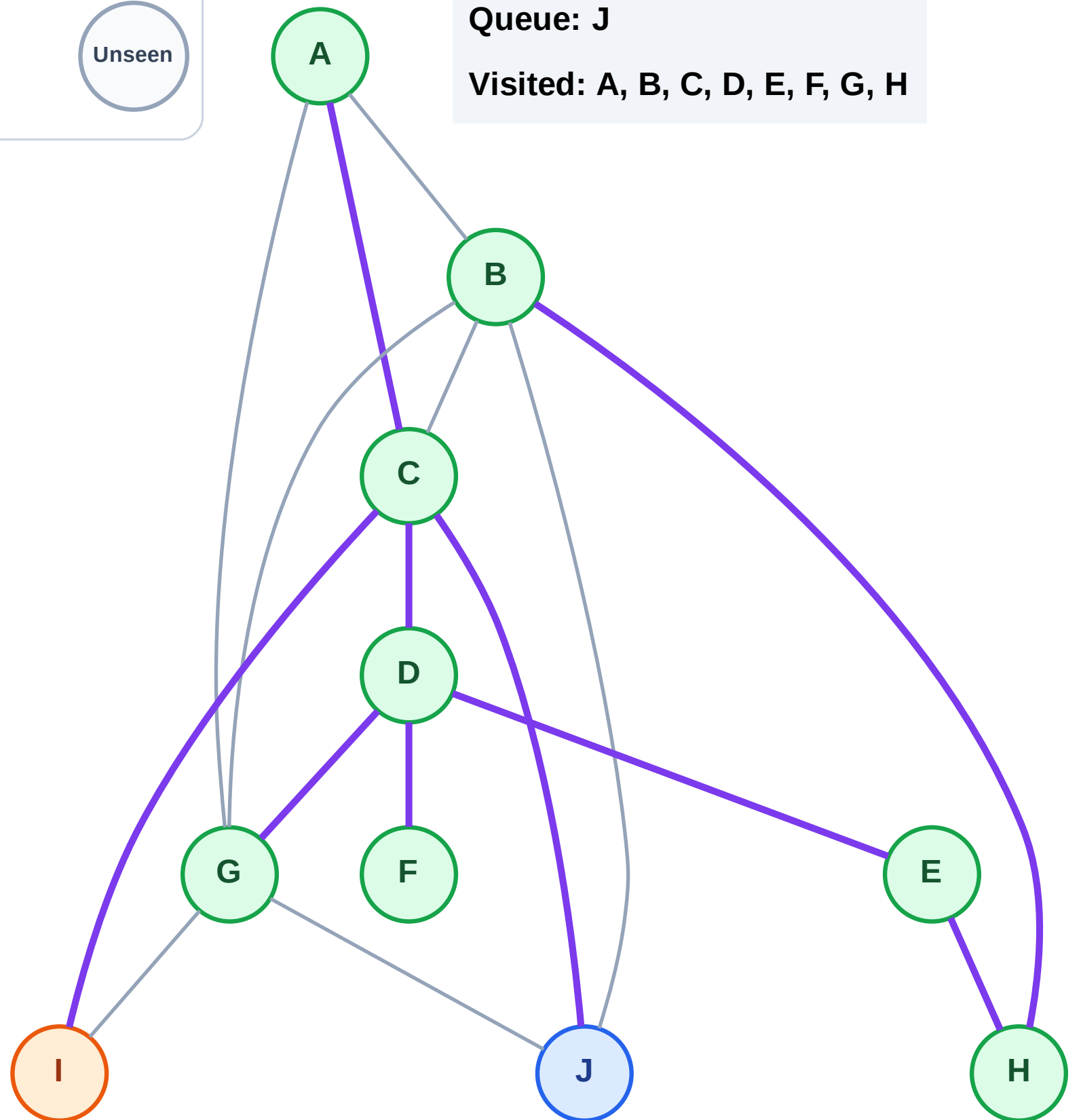
Step 18: Process node I

## Legend



Queue: J

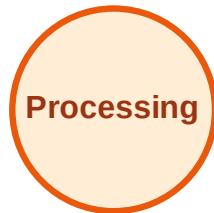
Visited: A, B, C, D, E, F, G, H



# BFS Traversal

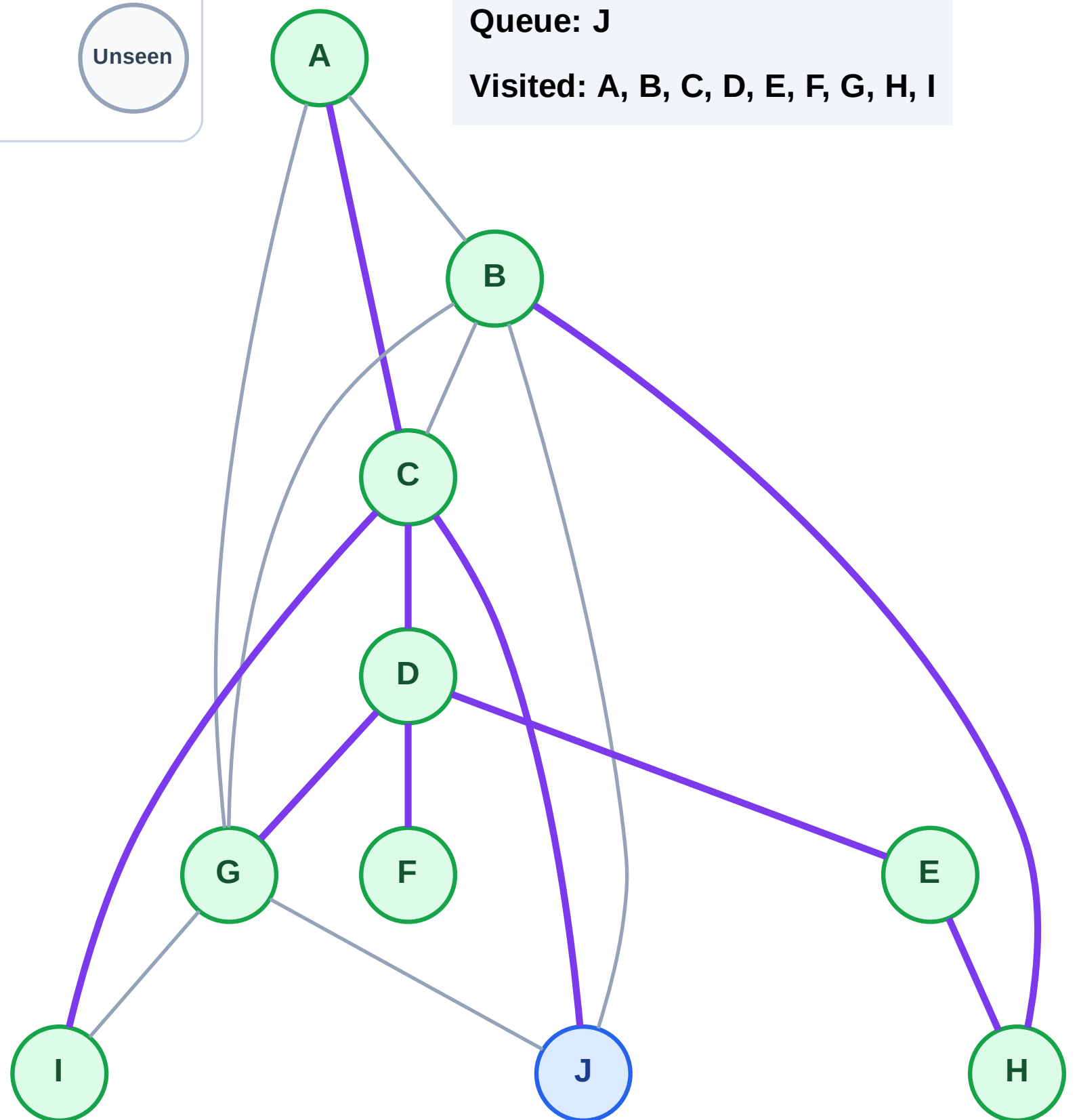
Step 19: No new neighbors from I

## Legend



Queue: J

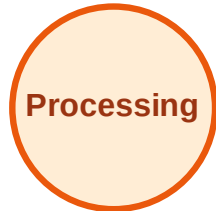
Visited: A, B, C, D, E, F, G, H, I



# BFS Traversal

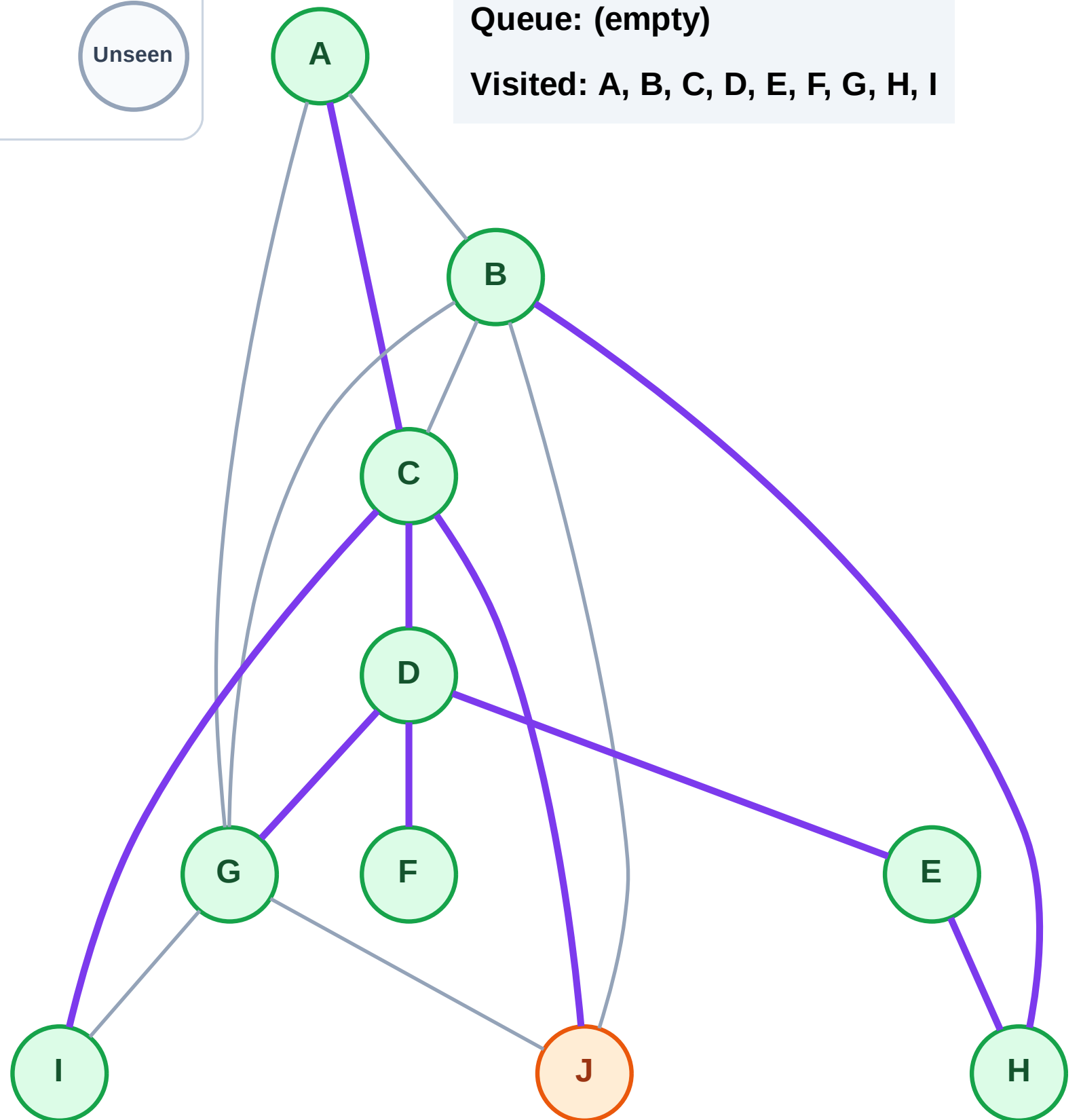
Step 20: Process node J

## Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I

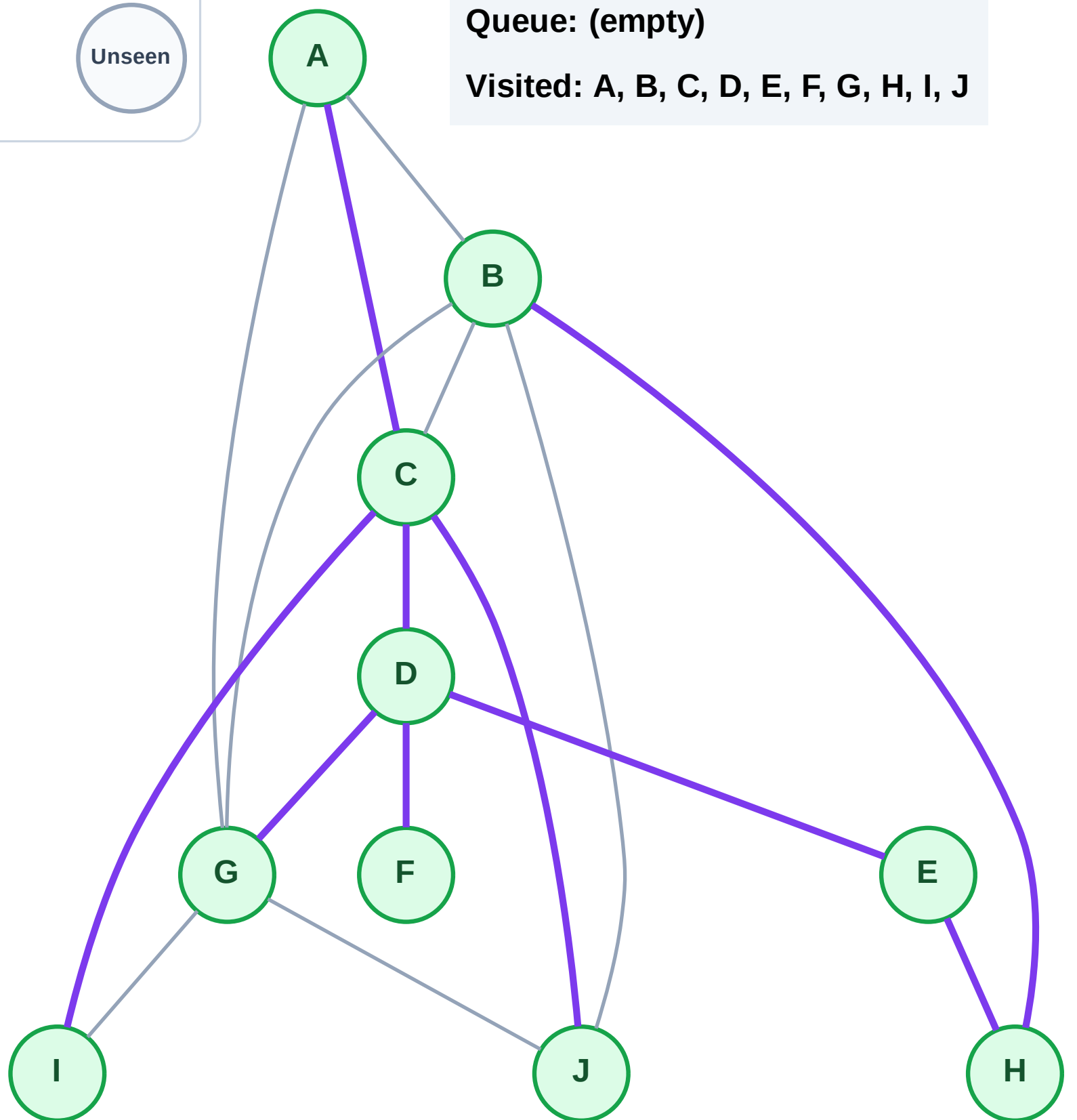


Step 21: No new neighbors from J

### Legend

## Unseen

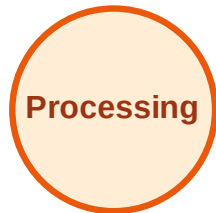
**Visited: A, B, C, D, E, F, G, H, I, J**



# DFS Traversal

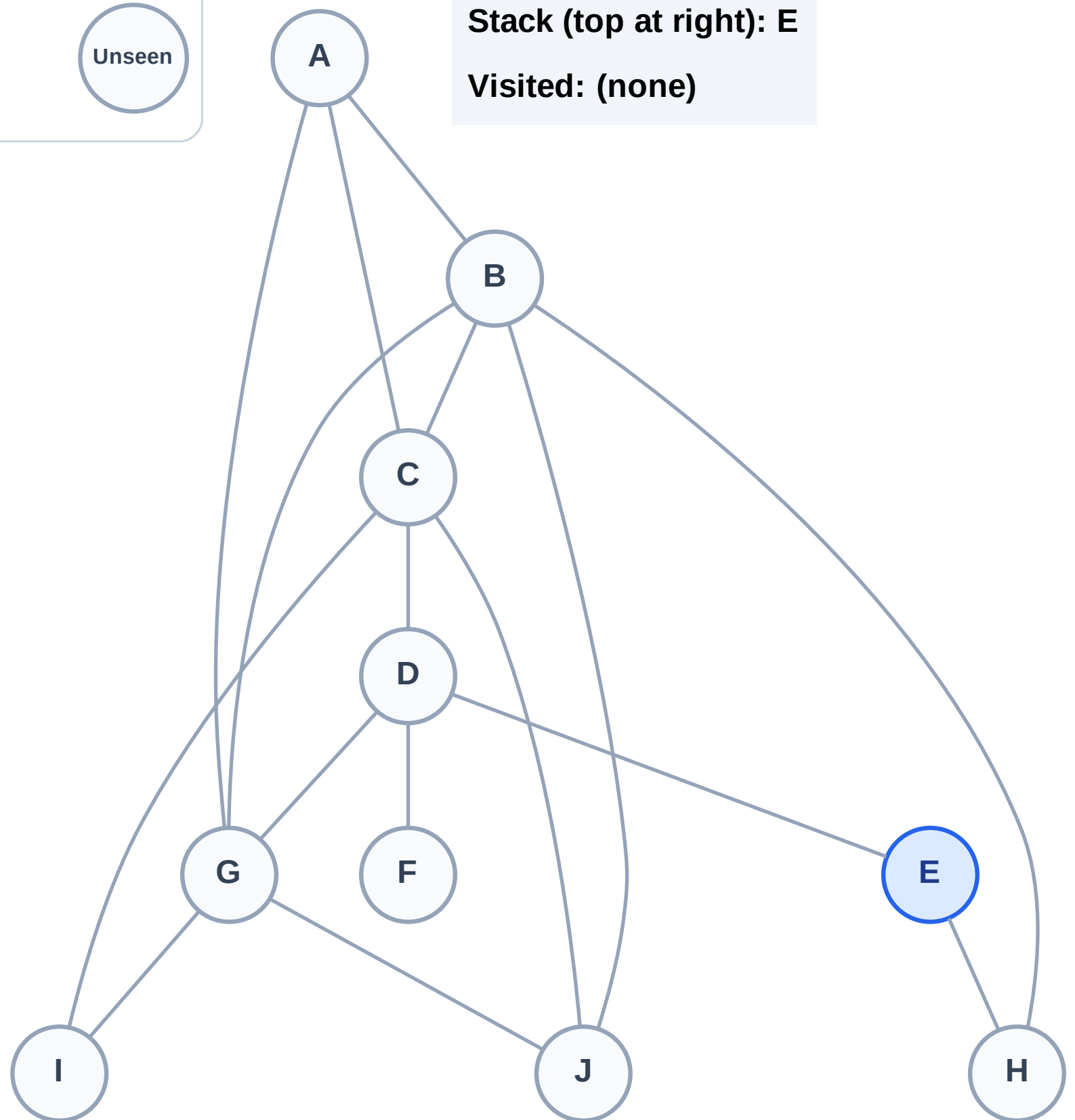
Step 1: Start at E

## Legend



Stack (top at right): E

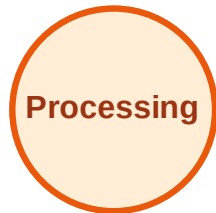
Visited: (none)



# DFS Traversal

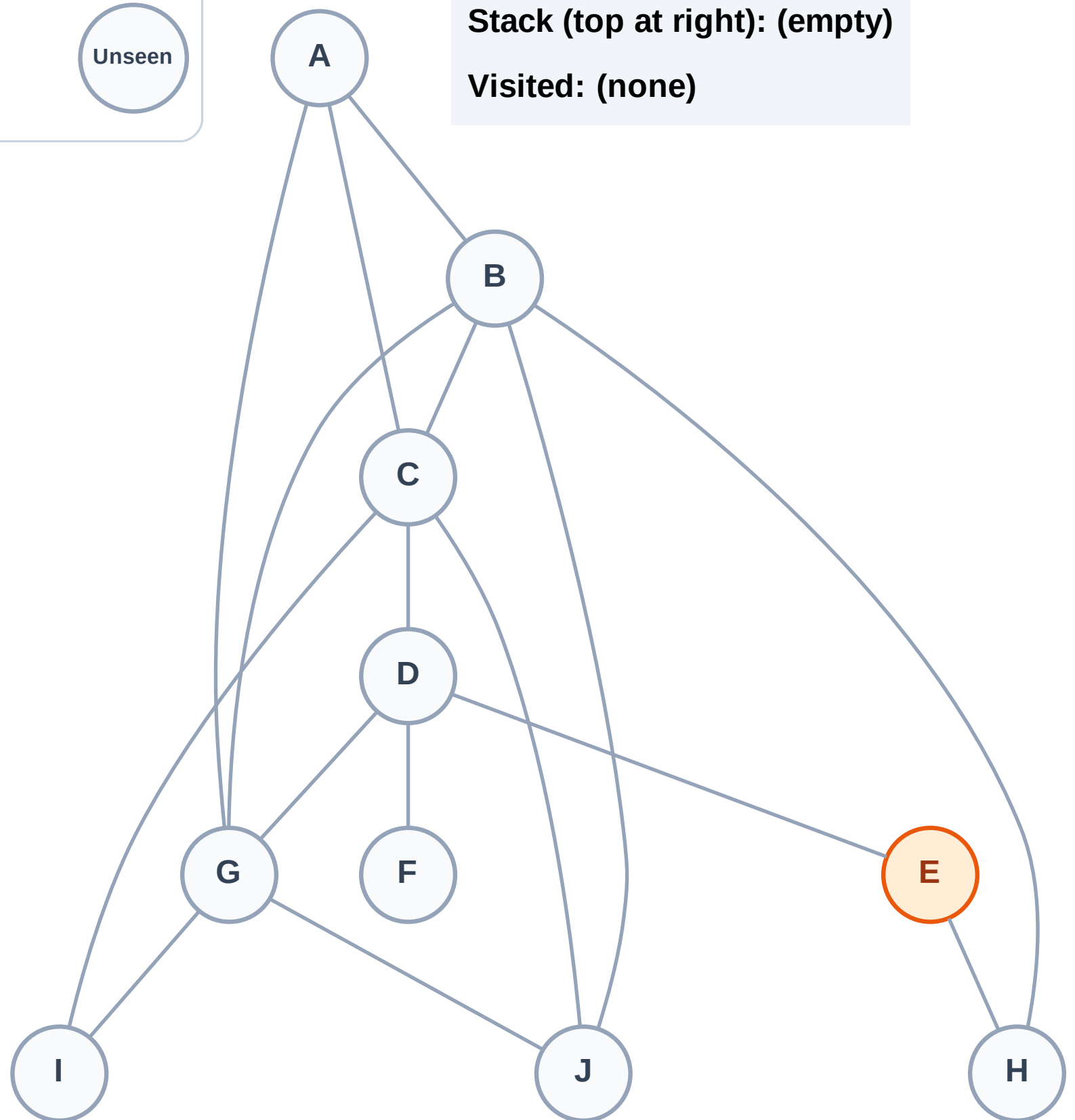
Step 2: Process node E

## Legend



Stack (top at right): (empty)

Visited: (none)

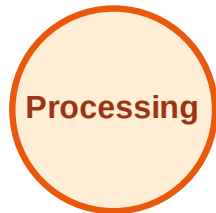




# DFS Traversal

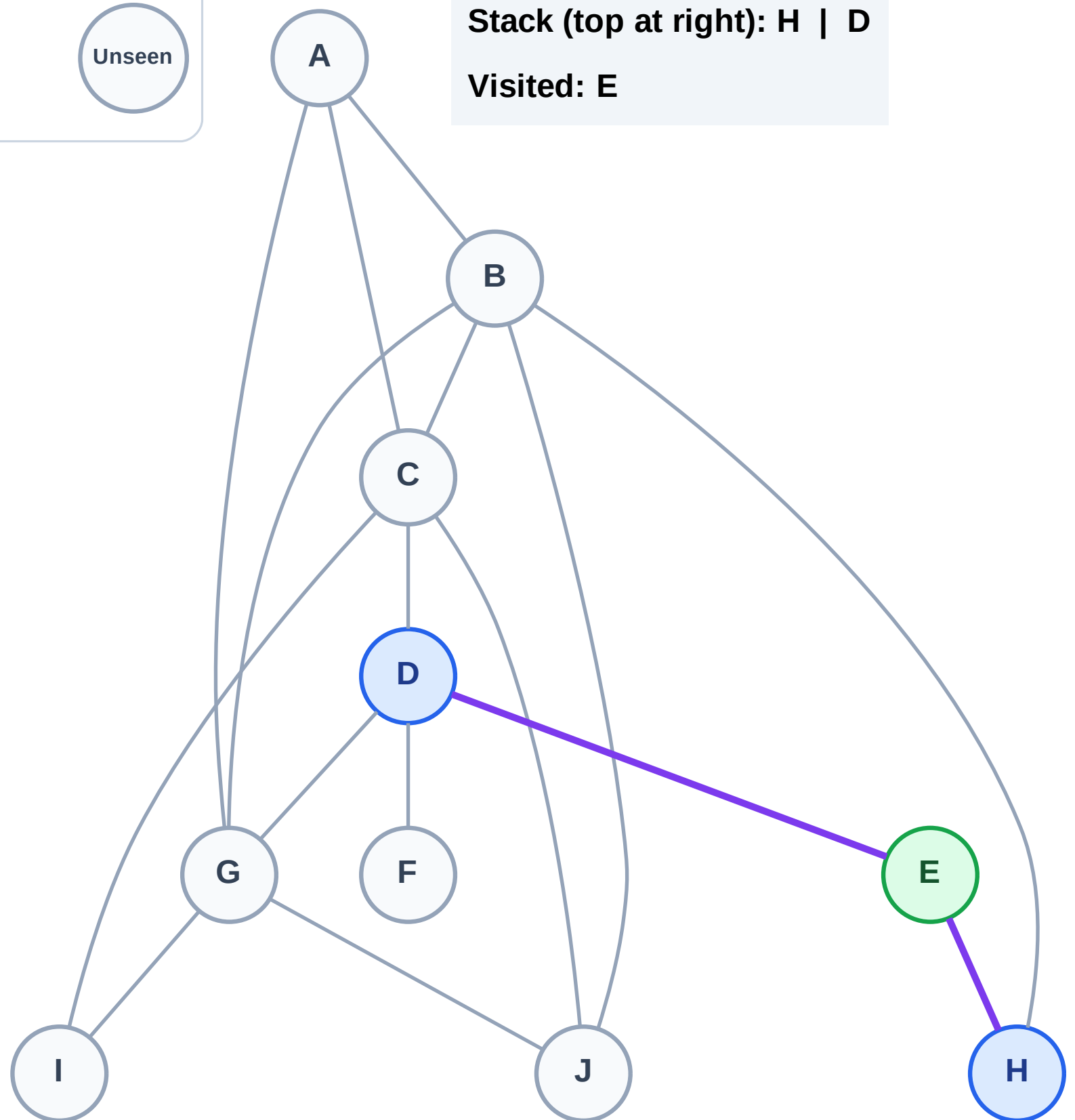
Step 3: Discover H, D from E

## Legend



Stack (top at right): H | D

Visited: E

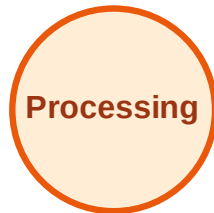




# DFS Traversal

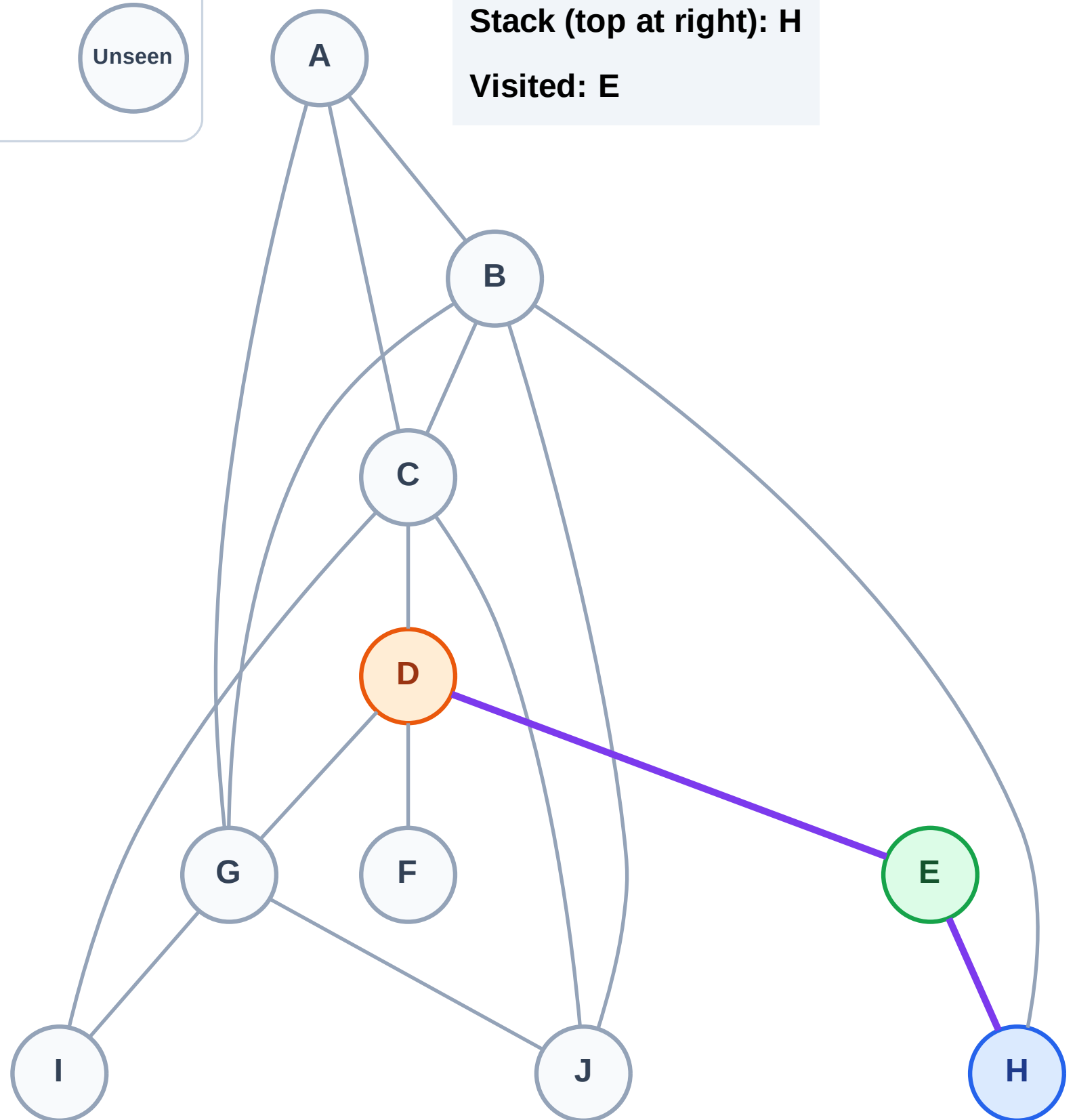
Step 4: Process node D

## Legend



Stack (top at right): H

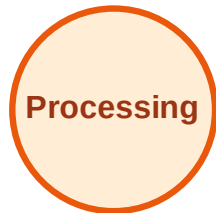
Visited: E



# DFS Traversal

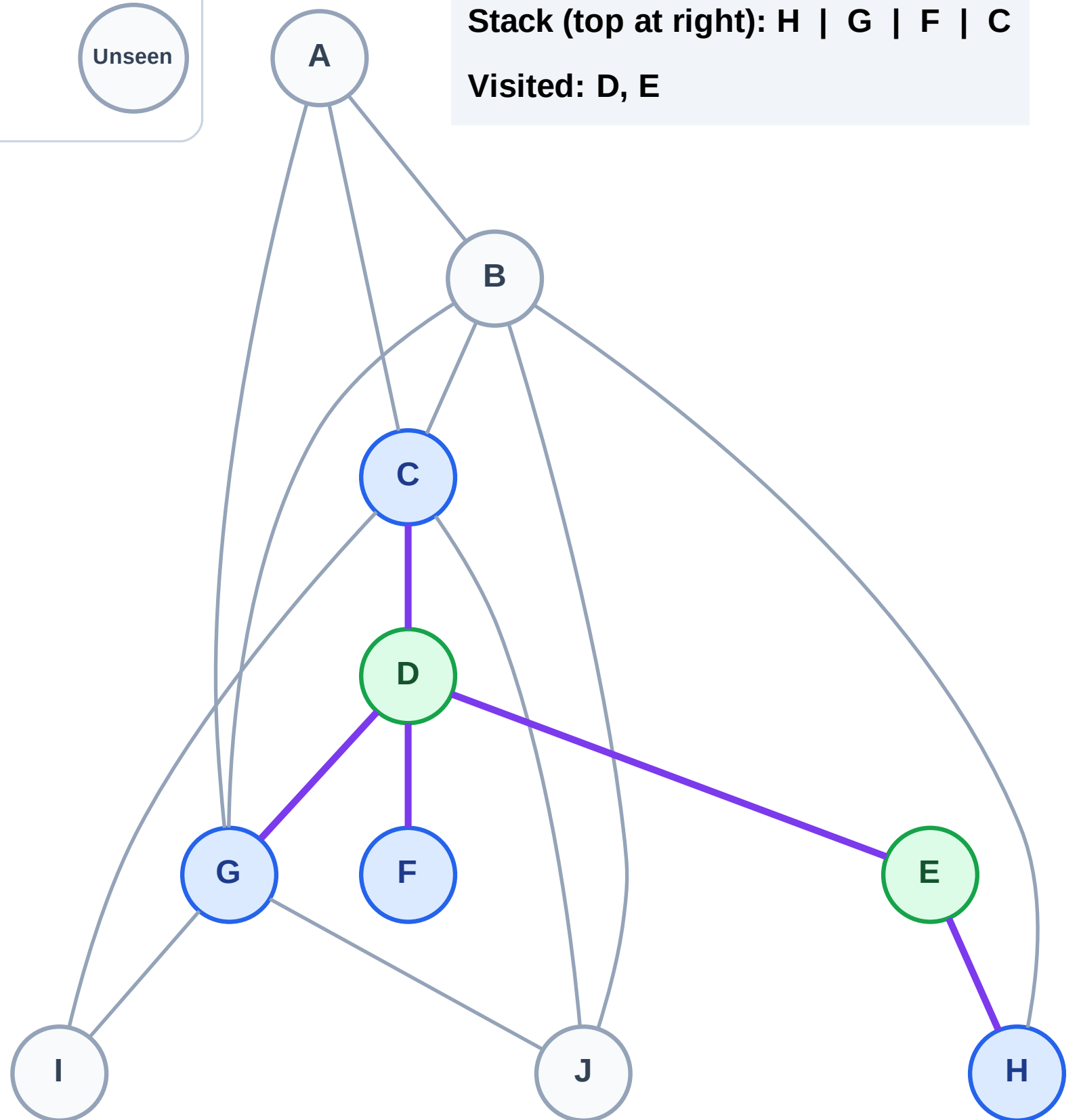
Step 5: Discover G, F, C from D

## Legend



Stack (top at right): H | G | F | C

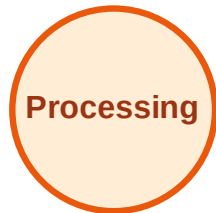
Visited: D, E



# DFS Traversal

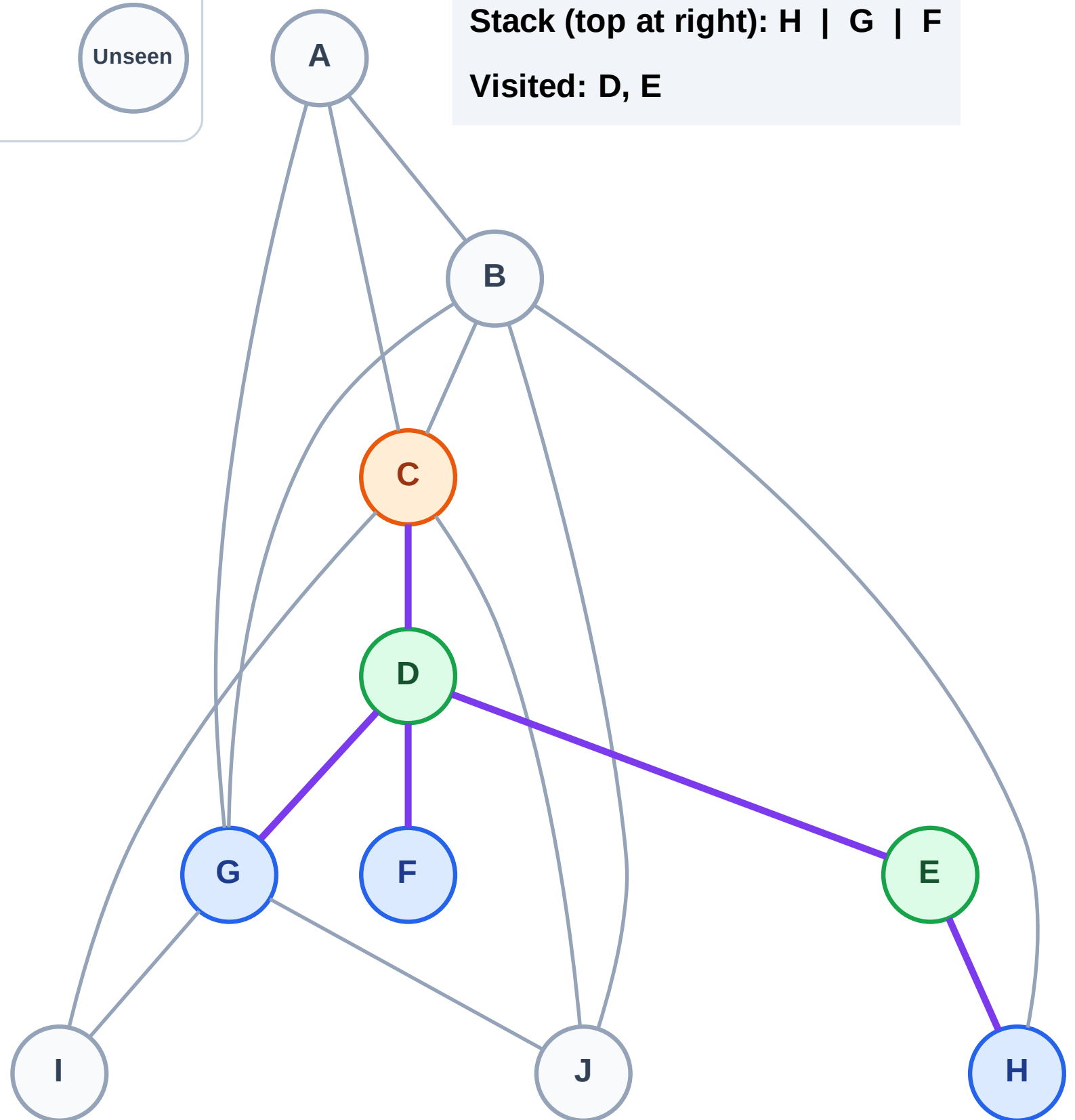
Step 6: Process node C

## Legend



Stack (top at right): H | G | F

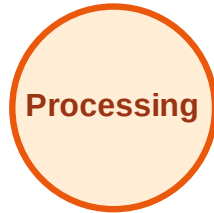
Visited: D, E



# DFS Traversal

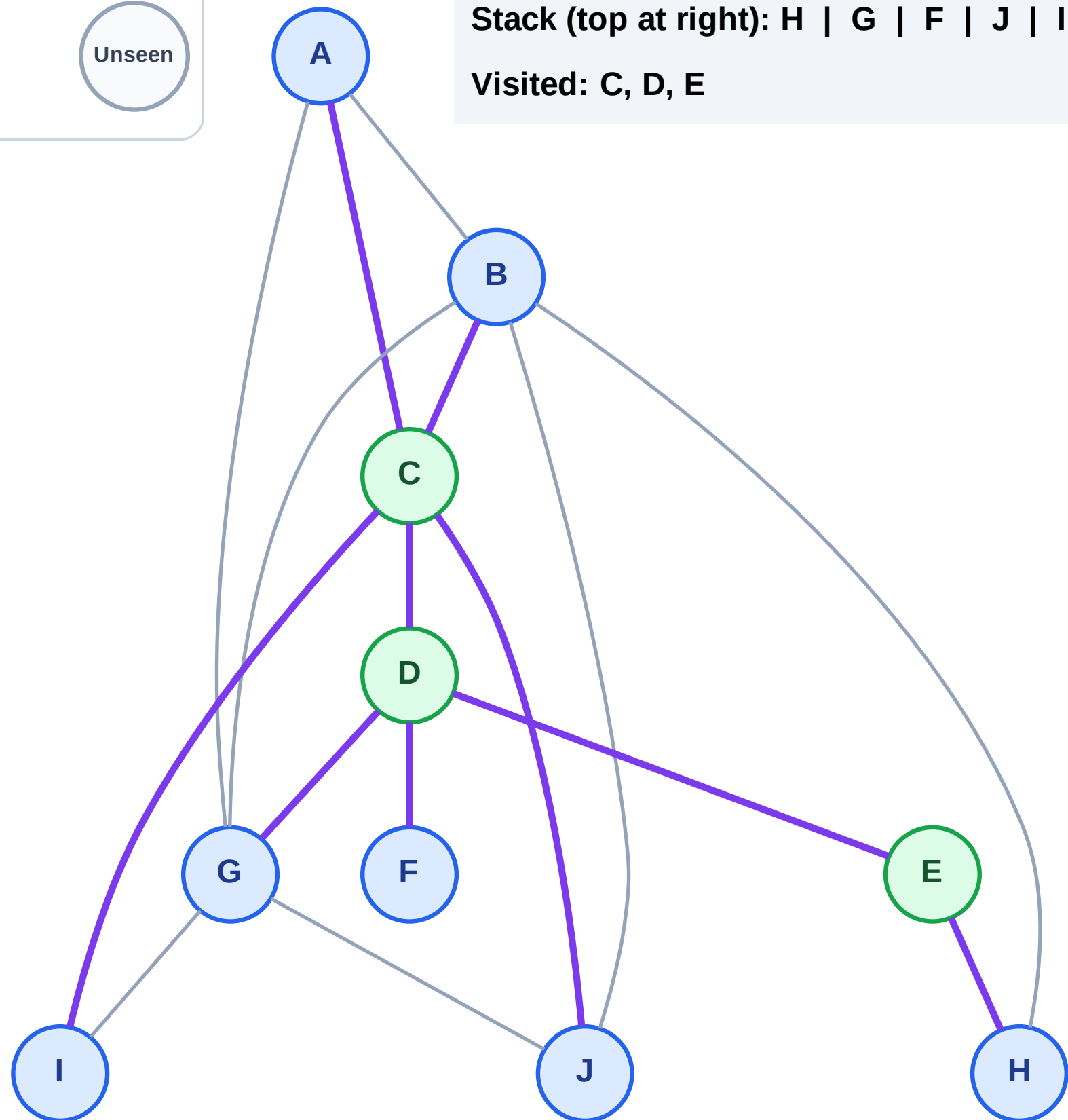
Step 7: Discover J, I, B, A from C

## Legend



Stack (top at right): H | G | F | J | I | B | A

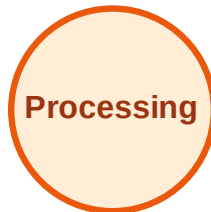
Visited: C, D, E



# DFS Traversal

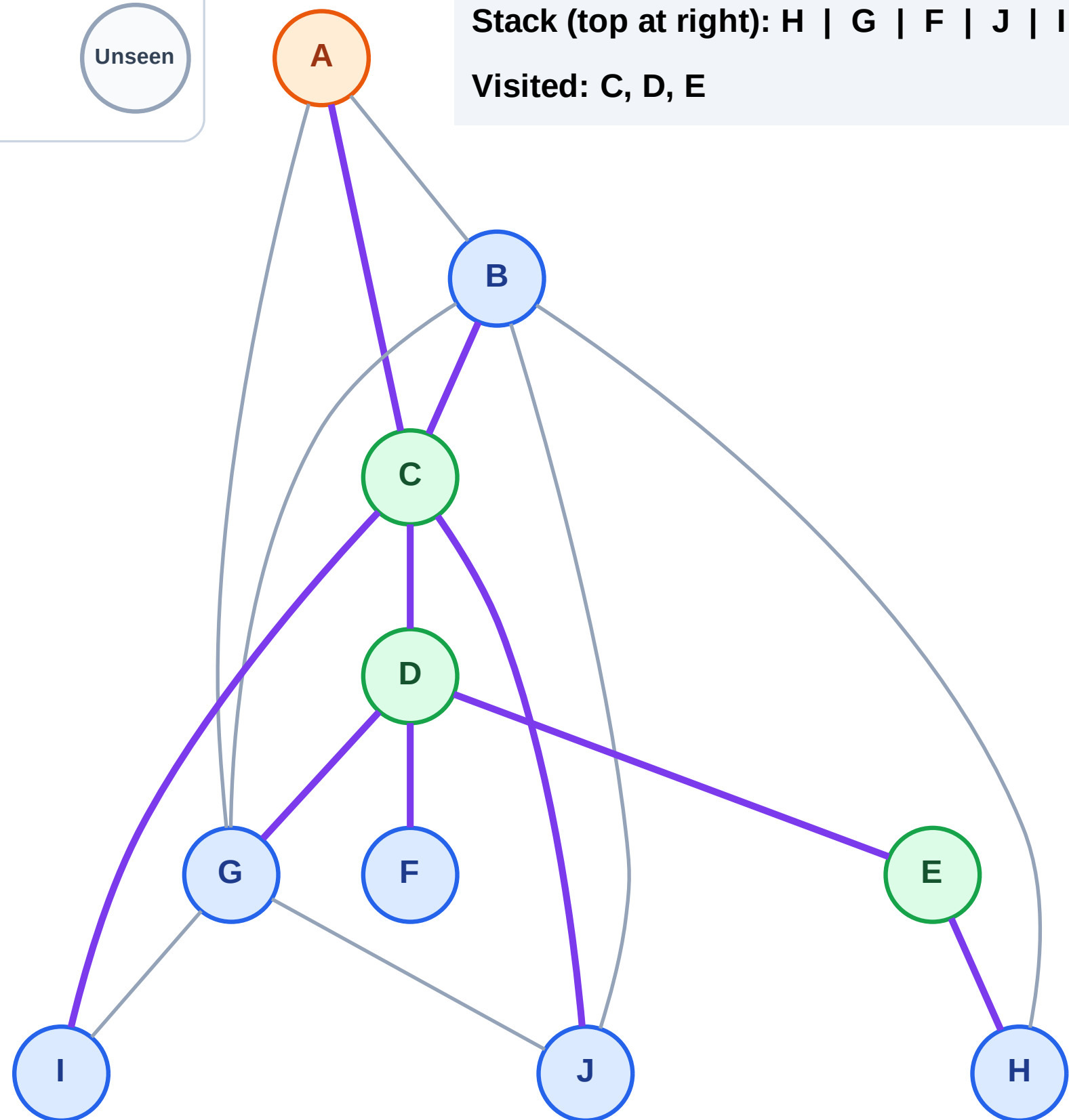
Step 8: Process node A

## Legend



Stack (top at right): H | G | F | J | I | B

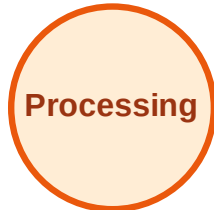
Visited: C, D, E



# DFS Traversal

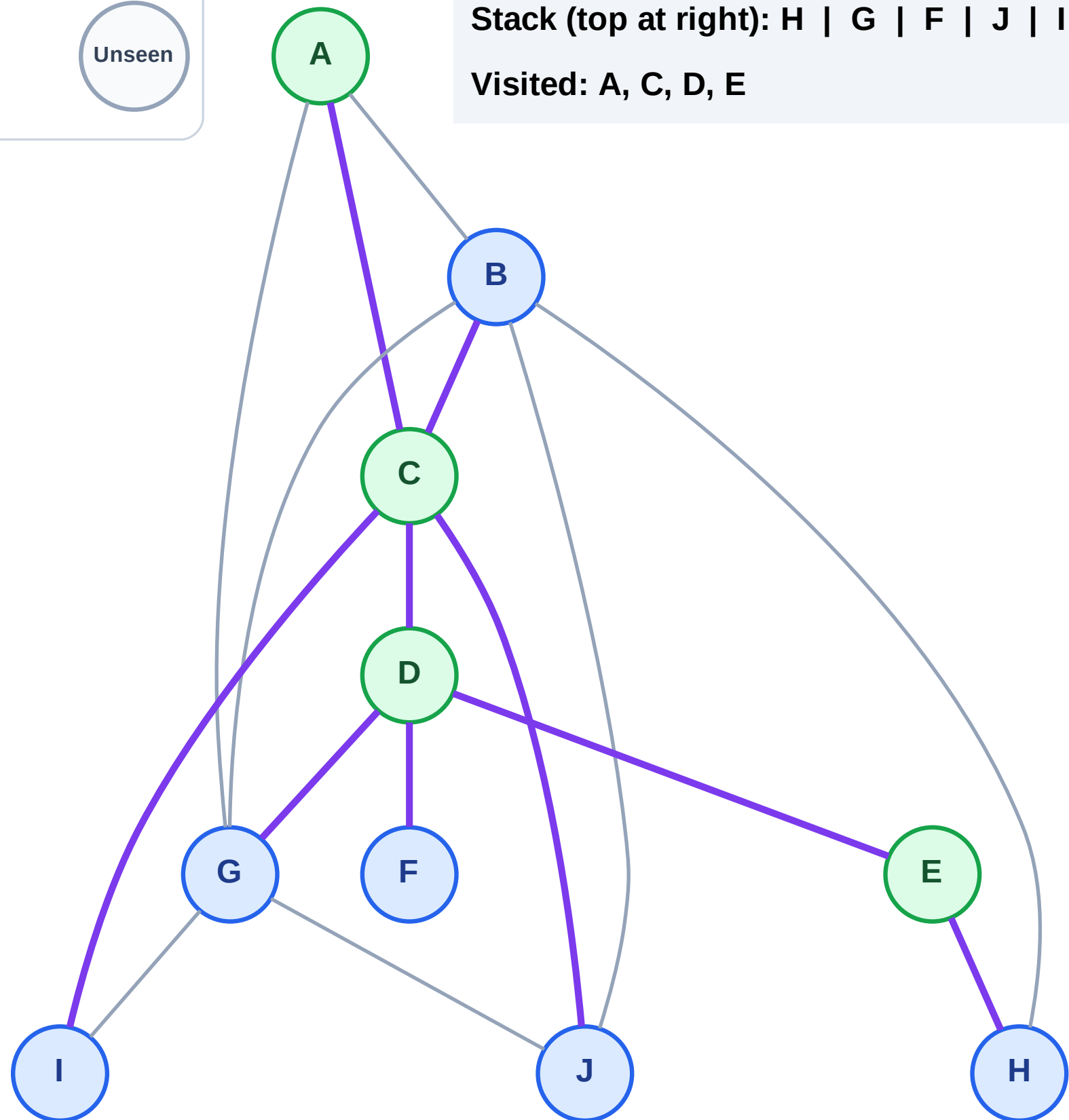
Step 9: No new neighbors from A

## Legend



Stack (top at right): H | G | F | J | I | B

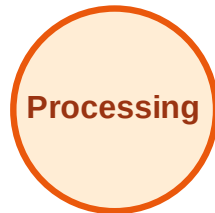
Visited: A, C, D, E



# DFS Traversal

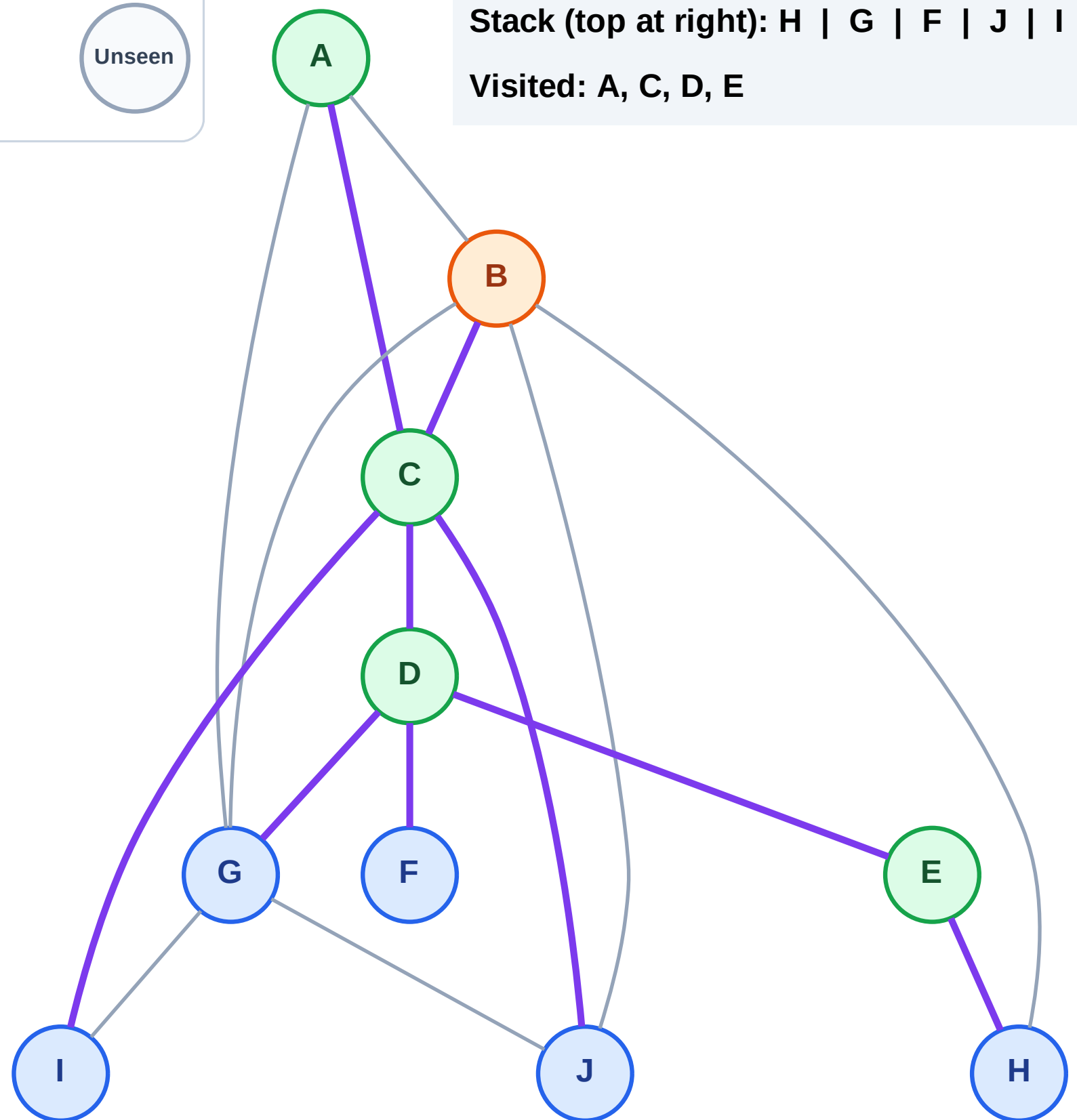
Step 10: Process node B

## Legend



Stack (top at right): H | G | F | J | I

Visited: A, C, D, E

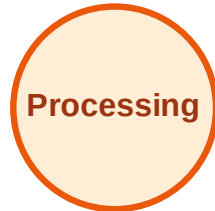




# DFS Traversal

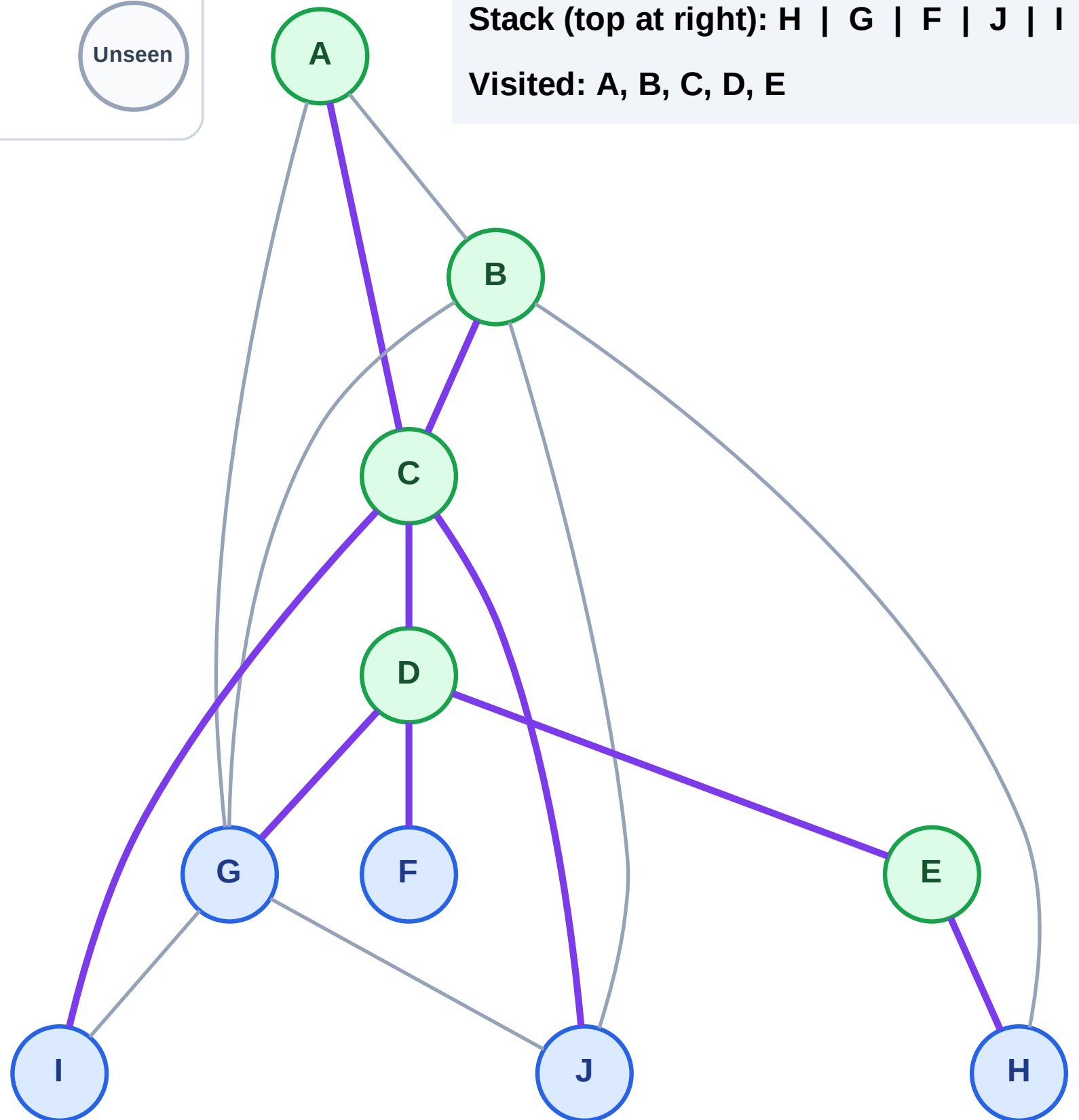
Step 11: No new neighbors from B

## Legend



Stack (top at right): H | G | F | J | I

Visited: A, B, C, D, E





# DFS Traversal

Step 12: Process node I

## Legend

Complete

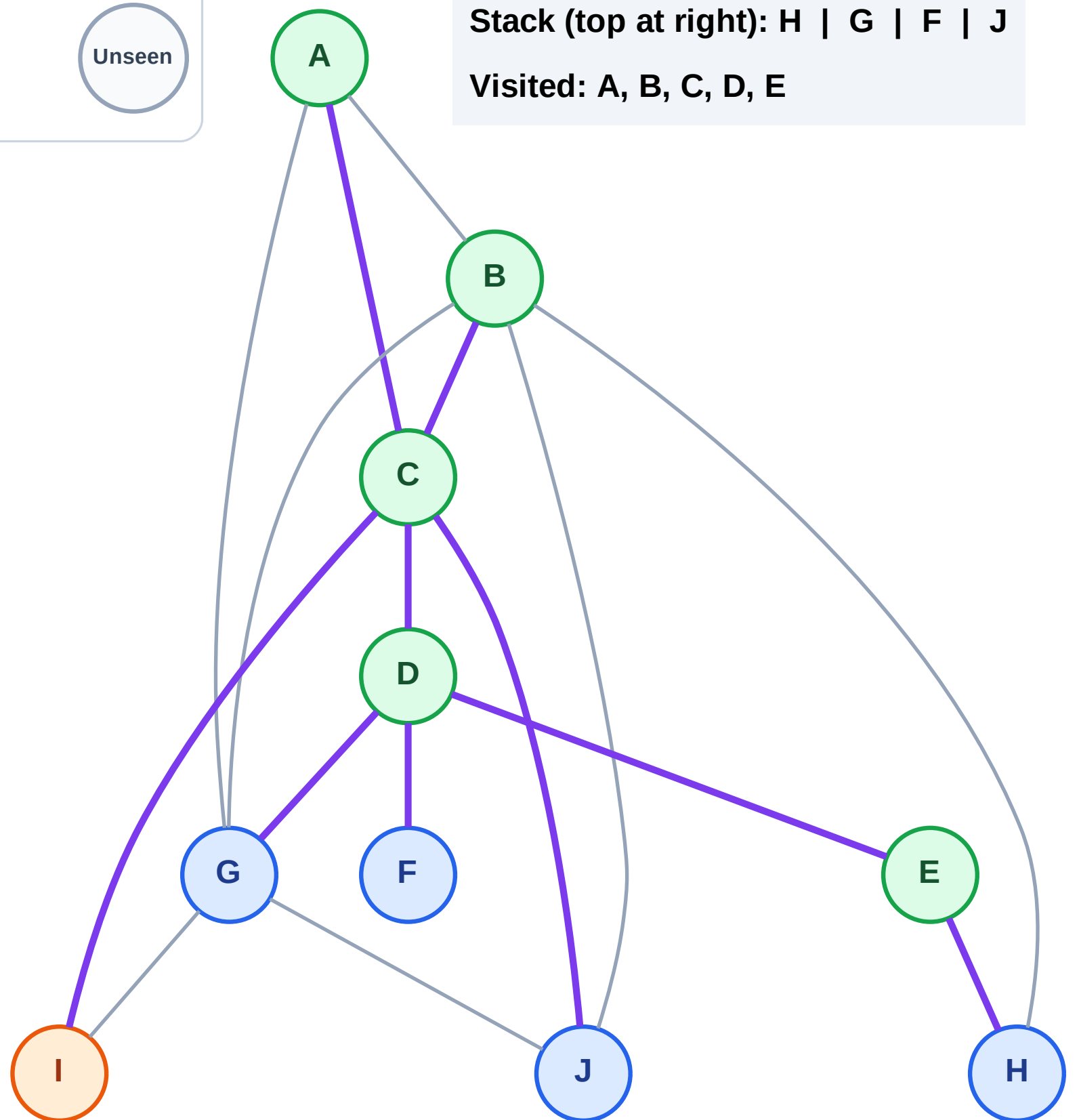
Processing

Discovered

Unseen

Stack (top at right): H | G | F | J

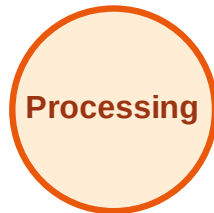
Visited: A, B, C, D, E



# DFS Traversal

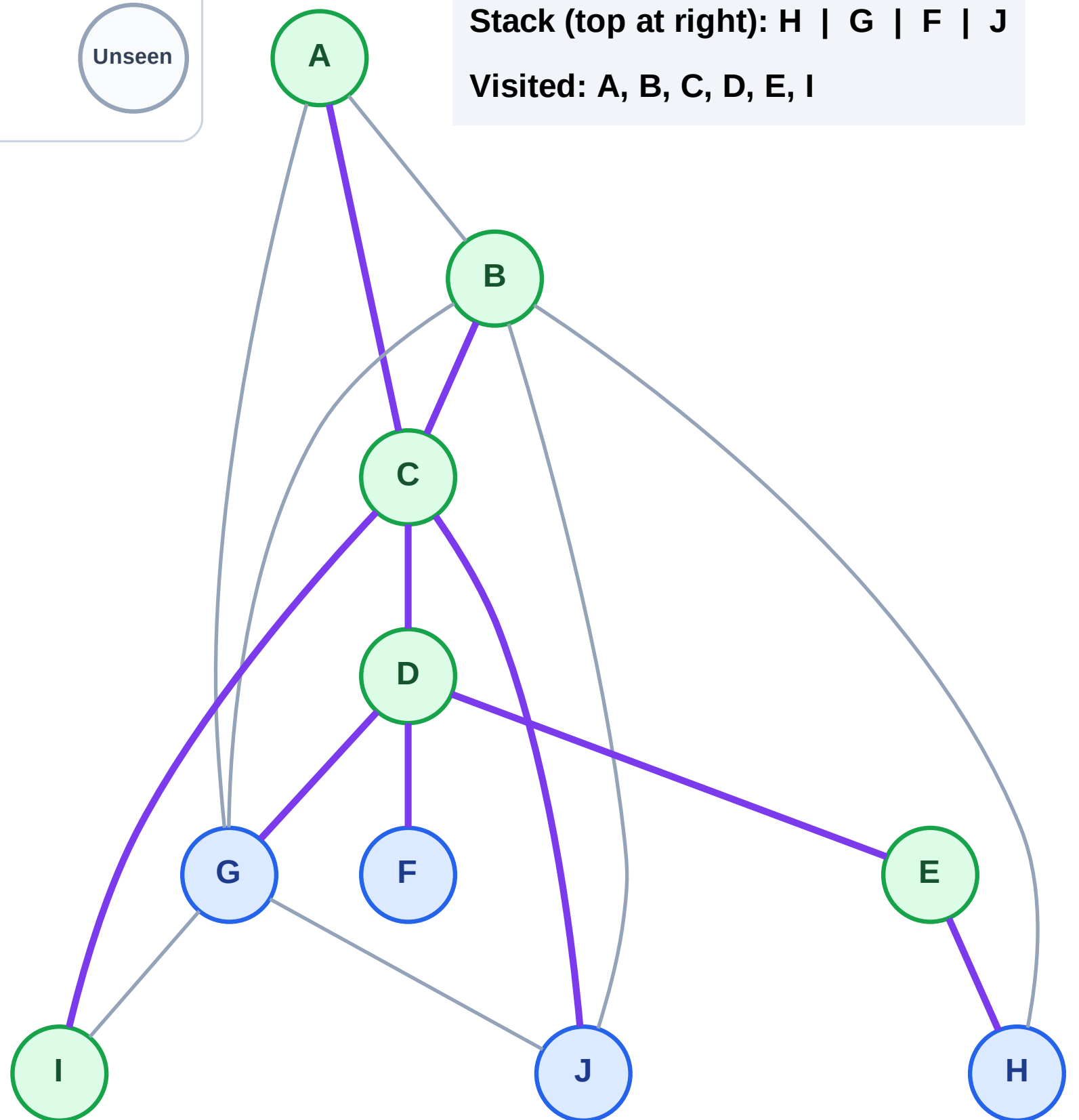
Step 13: No new neighbors from I

## Legend



Stack (top at right): H | G | F | J

Visited: A, B, C, D, E, I



# DFS Traversal

Step 14: Process node J

Legend

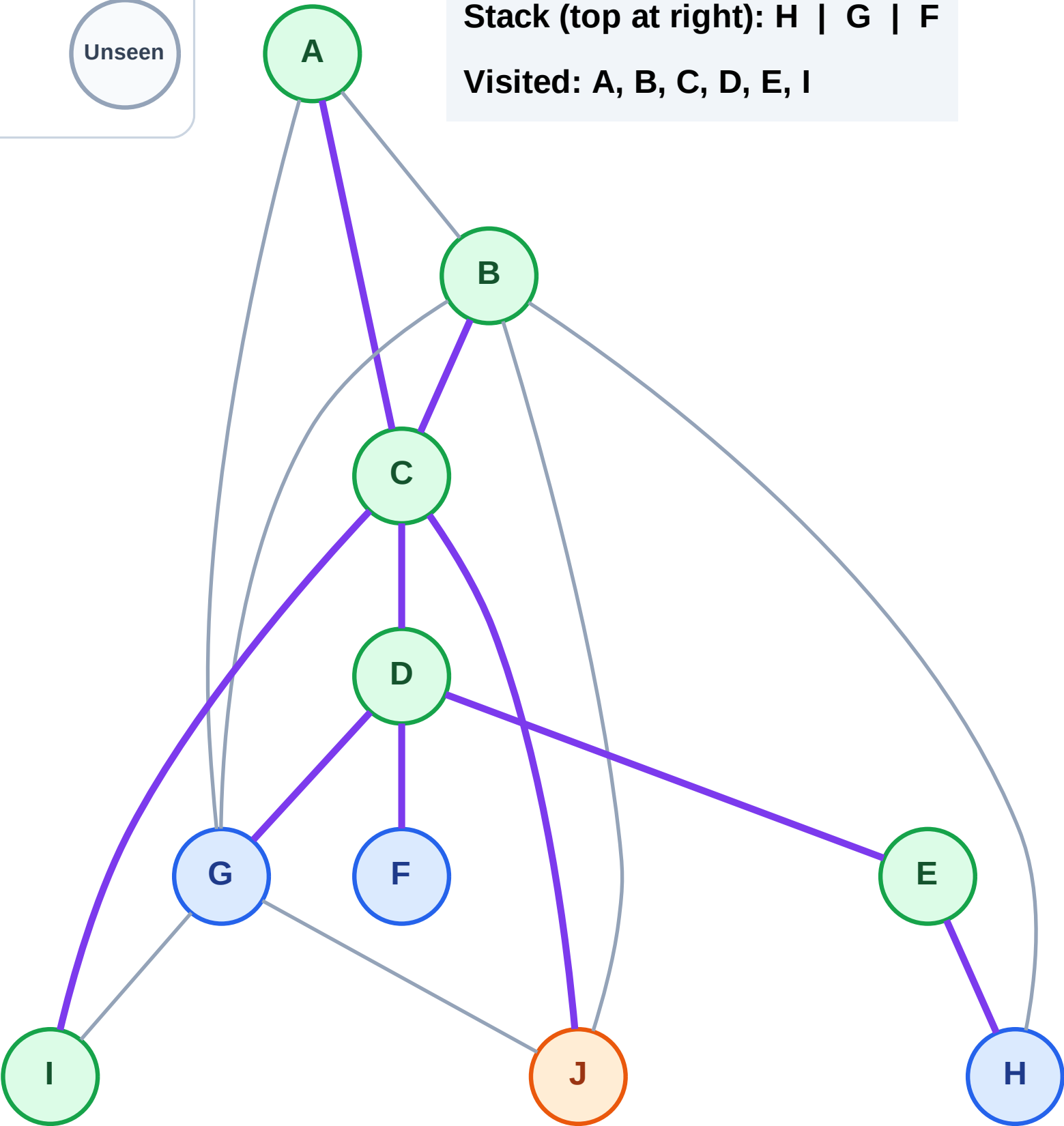
Complete

Processing

Discovered

Unseen

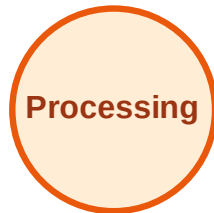
Stack (top at right): H | G | F  
Visited: A, B, C, D, E, I



# DFS Traversal

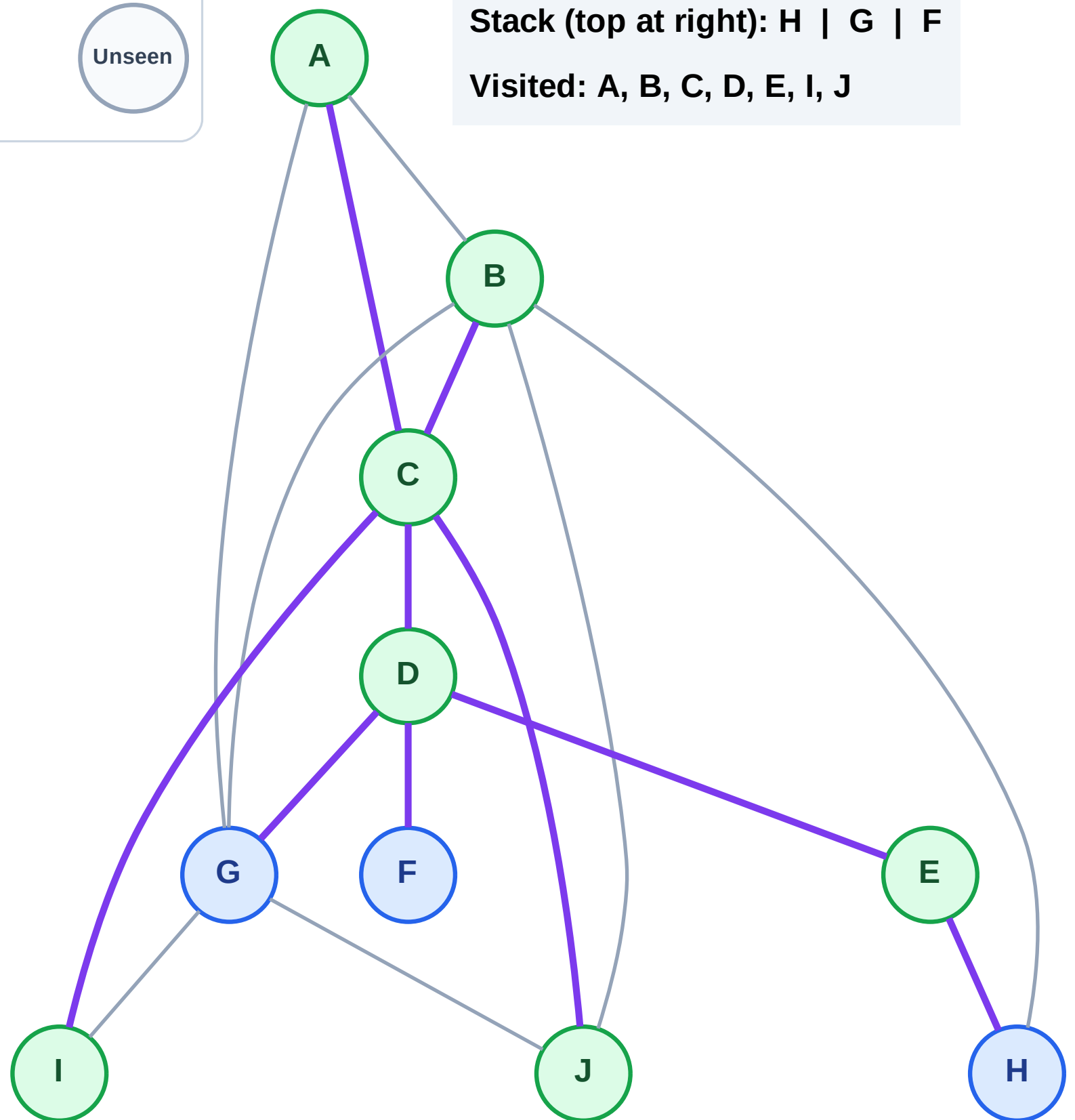
Step 15: No new neighbors from J

## Legend



Stack (top at right): H | G | F

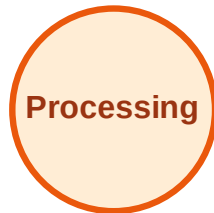
Visited: A, B, C, D, E, I, J



# DFS Traversal

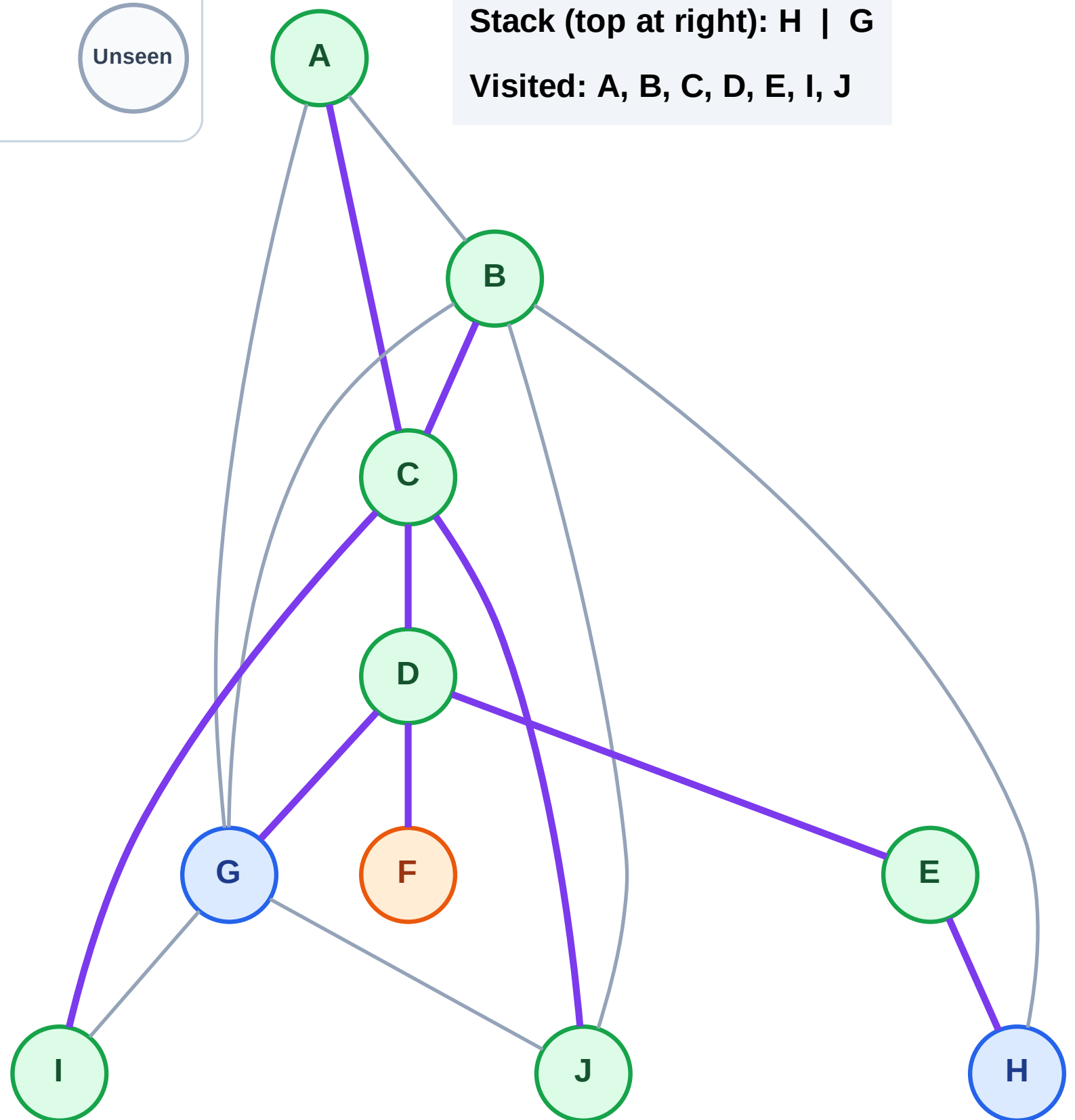
Step 16: Process node F

## Legend



Stack (top at right): H | G

Visited: A, B, C, D, E, I, J



# DFS Traversal

Step 17: No new neighbors from F

## Legend

Complete

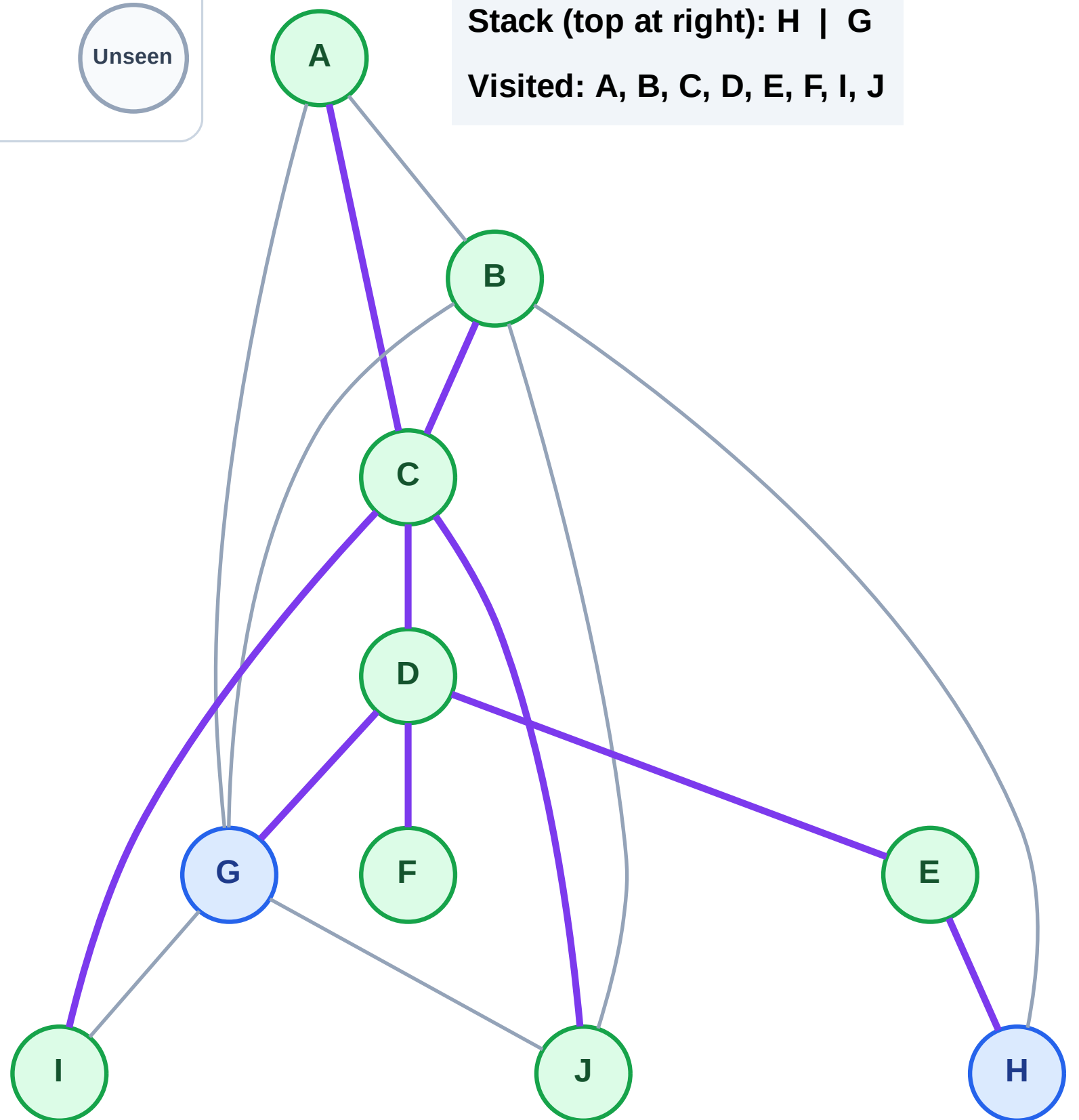
Processing

Discovered

Unseen

Stack (top at right): H | G

Visited: A, B, C, D, E, F, I, J



# DFS Traversal

Step 18: Process node G

## Legend

Complete

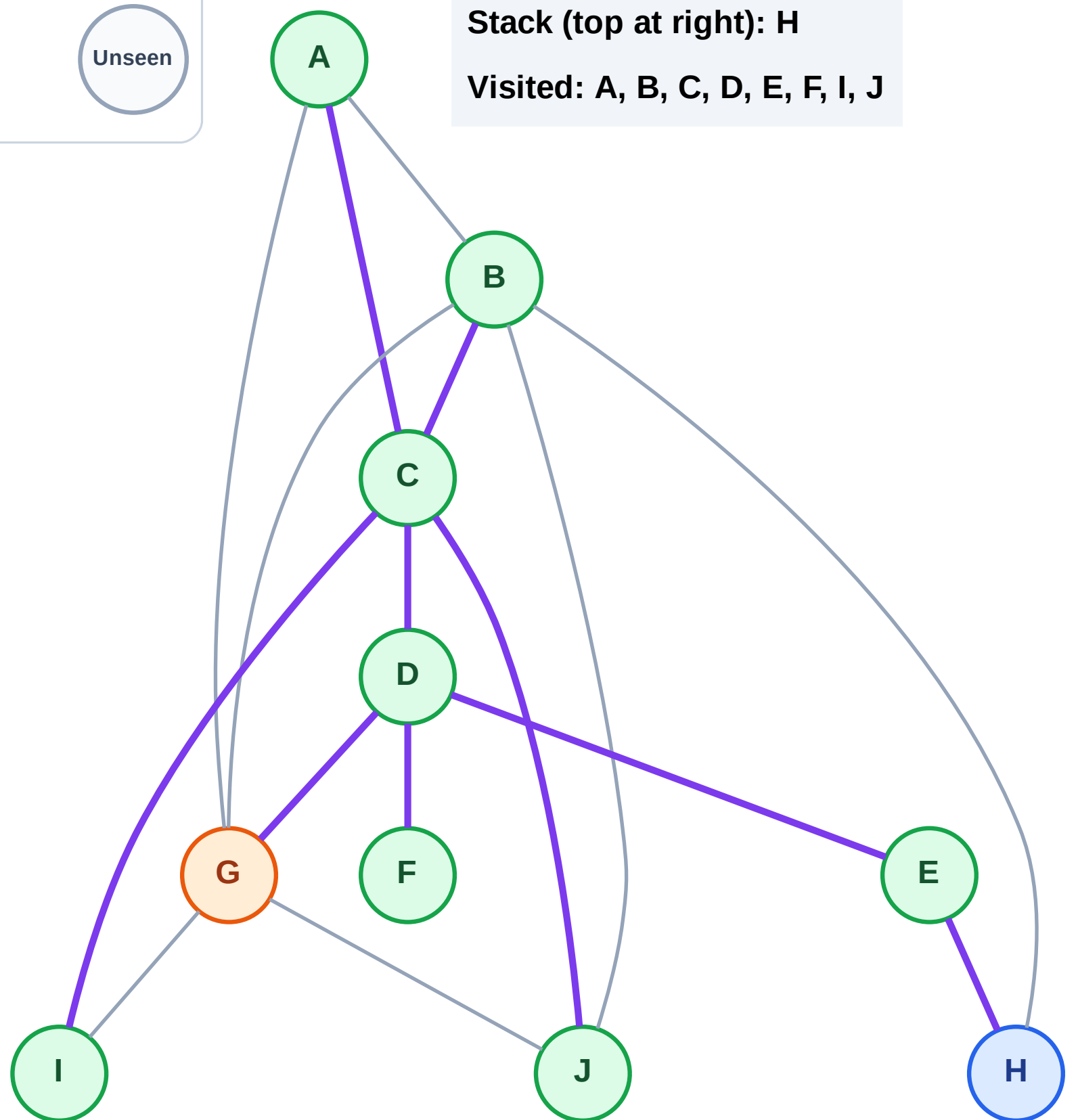
Processing

Discovered

Unseen

Stack (top at right): H

Visited: A, B, C, D, E, F, I, J

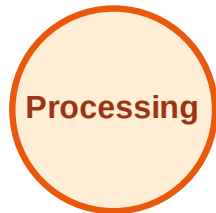




# DFS Traversal

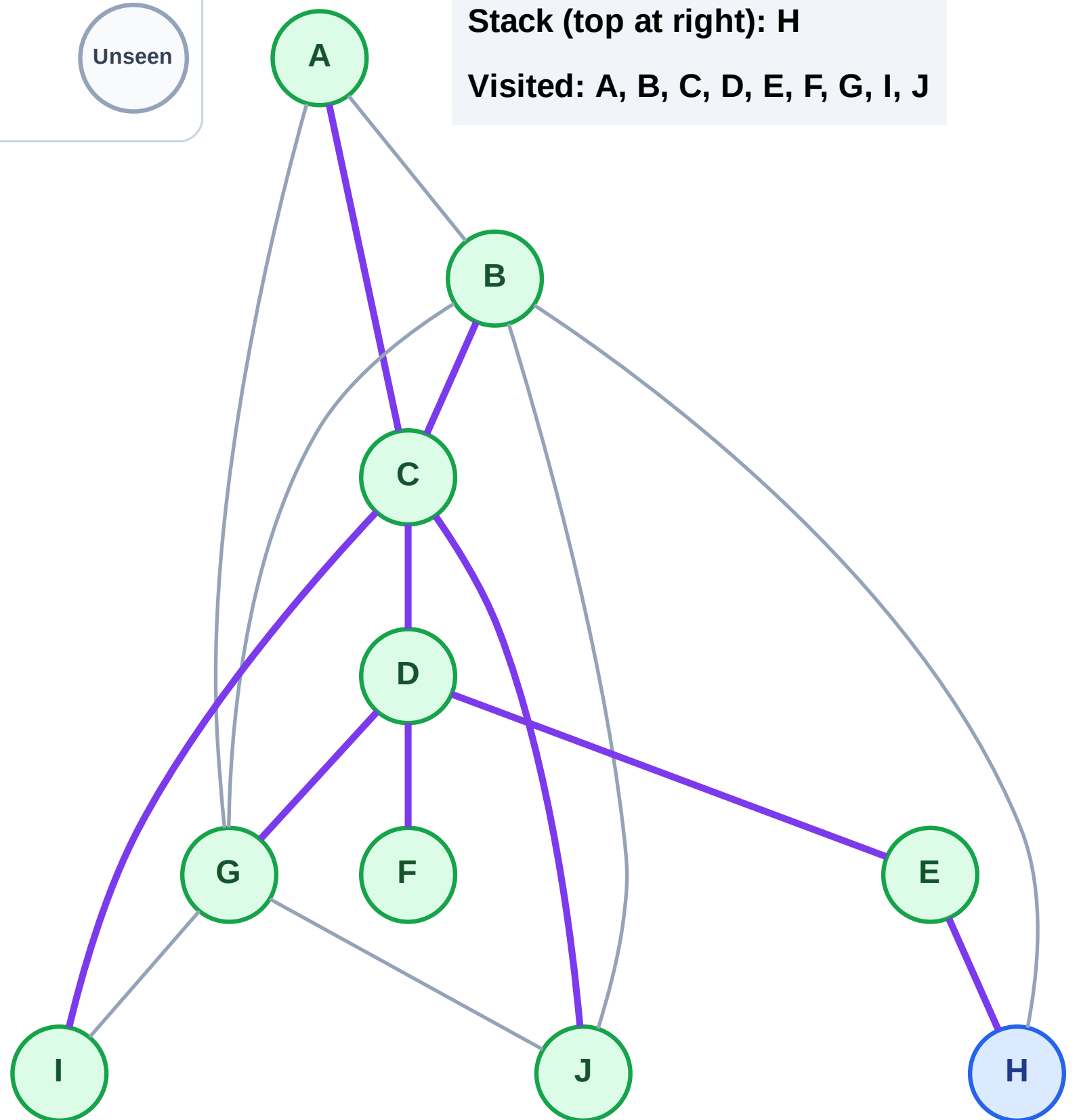
Step 19: No new neighbors from G

## Legend



Stack (top at right): H

Visited: A, B, C, D, E, F, G, I, J

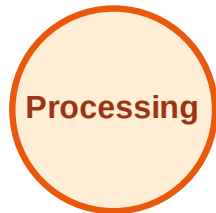




# DFS Traversal

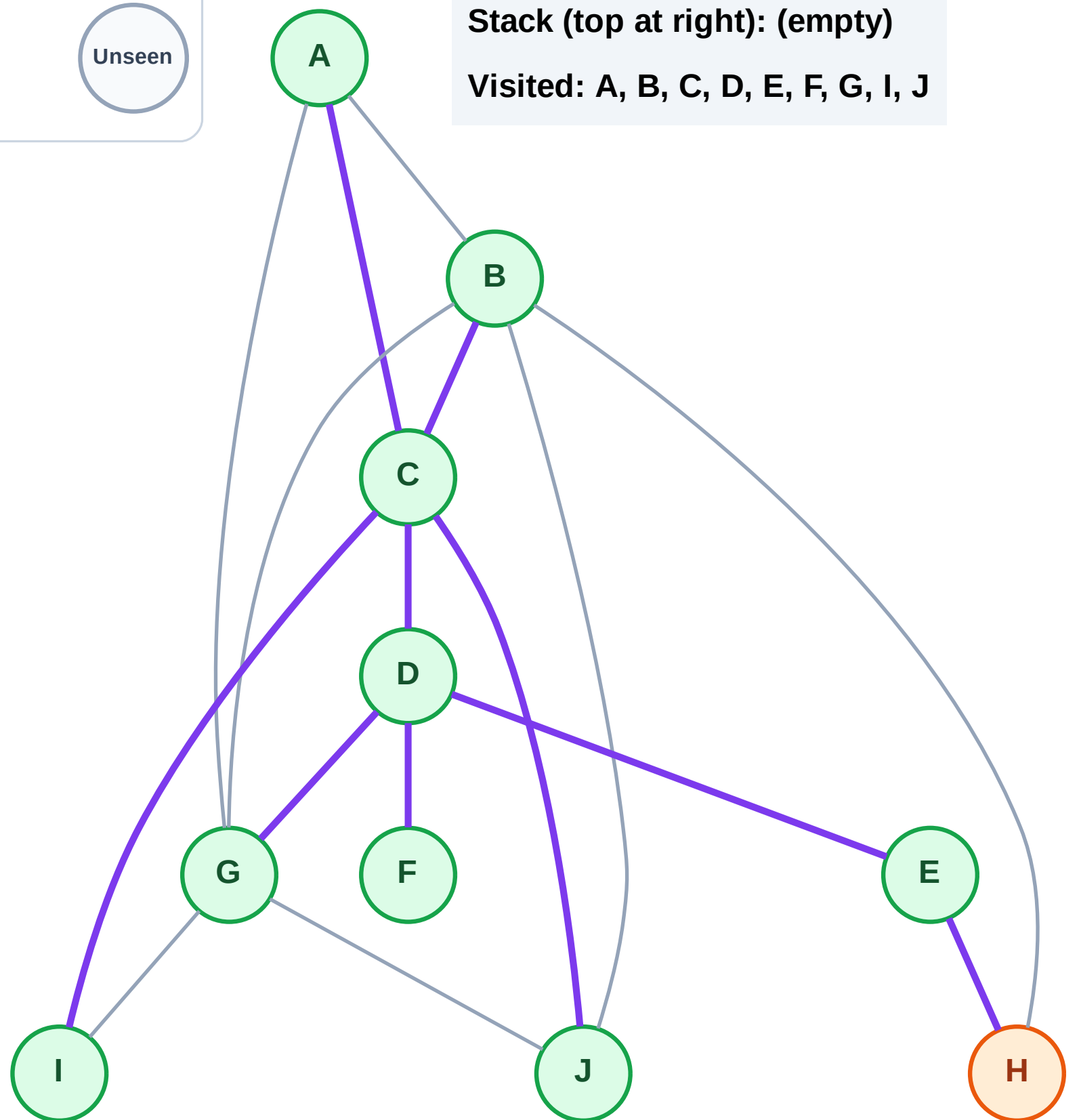
Step 20: Process node H

## Legend



Stack (top at right): (empty)

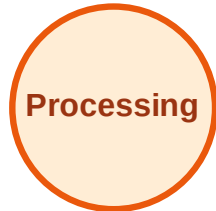
Visited: A, B, C, D, E, F, G, I, J



# DFS Traversal

Step 21: No new neighbors from H

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

